



EXOSKELETON WALKER

PRODUCT & SERVICE SPACE DESIGN

This project aims to improve walking aids and medical spaces for people with mobility difficulties. Canes are crucial for those with leg and foot disabilities, providing support for the elderly and patients, but they can be inconvenient to carry. Inspired by my grandfather, who faced challenges using a cane for medical reasons, I undertook this project to help others in similar situations.

KEYWORDS Walking assistance, Stability walking, Injury care, Exoskeleton, Intelligent positioning, Service assistance

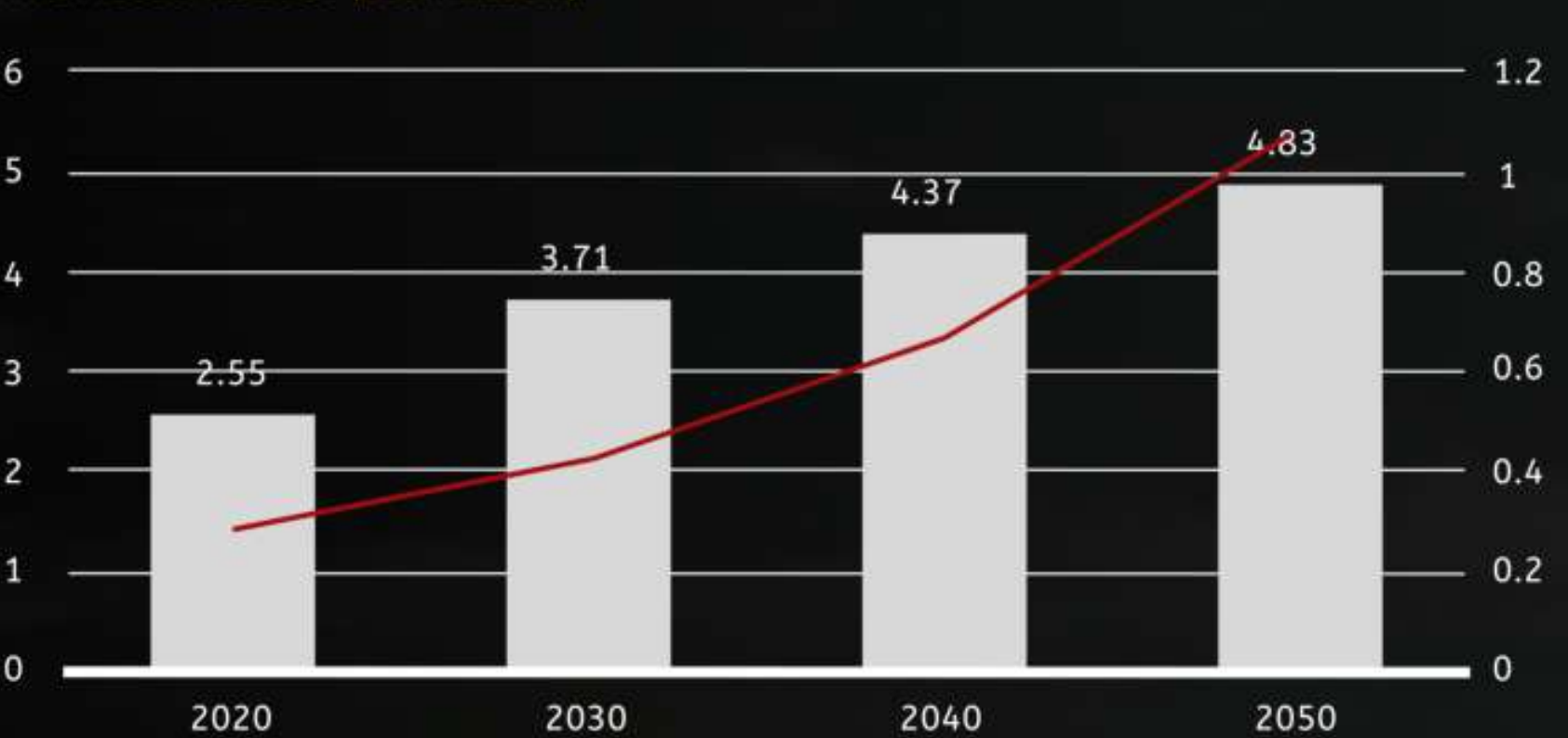
MATERIAL PLA 3D printing, Velcro, Gyroscope, Vibration sensor, GPS, Temperature sensor

SIZE Product - Exoskeleton Walker 4CM*1.5CM*70CM
Space - Service Space 60M*3.5M*70M

BACKGROUND RESEARCH

In this project, my aim is to develop a novel assistive device akin to a cane. I have conducted research on the canes currently available in the market, gathering insights into their strengths and weaknesses. It is my aspiration that this project will effectively address these limitations and integrate as many advantages from different types of canes as feasible.

ESTIMATION OF CHINA'S ELDERLY POPULATION FROM 2020 TO 2050 (100 MILLION)



ESTIMATION OF CHINA'S ELDERLY POPULATION FROM 2020 TO 2050 (100 MILLION)



COMMON CANE MATERIALS

Mostly carbon fiber, alloy materials, but also includes PVC and leather materials as an auxiliary.



TYPES OF CRUTCHES

There are many types of crutches that are commonly found on the market:

CANE

(1) T-cane:

It is a walking aid supported by one side of the hand, and the muscle strength of the upper arm and shoulder must be normal in order to maintain considerable stability when using the cane.



(2) Forearm support crutches.

The user fixes the forearm on top of the arm rest and uses the forearm to support the weight in order to assist walking.



(3) Axillary support crutches.

Standing with armpit support, forearms and hands are able to move around a bit.



DISADVANTAGES

- 1. Easy to lose.
- 2. Single crutches tend to cause the body to tilt.
- 3. Not convenient to carry.
- 4. The arm will be sore after long time use.

EXOSKELETON

The exoskeleton is more suitable for people who have been injured and rehabbed to return to normal walking quickly. Unlike crutches, which take up a lot of space and body parts, the exoskeleton simply fits over the joints.

DISADVANTAGES

- 1. Expensive
- 2. If it is purely electrically powered it can become a liability when there is no electricity.



WALKING FRAM

Structurally stable and easy to use. For using this support ability is very strong.

DISADVANTAGES

- 1. walkers are too conspicuous and most elderly people do not like to use them.
- 2. Heavy and not easy to carry.



MARKET RESEARCH

TARGET DEMOGRAPHIC:

- 1. Bad joints
- 2. Injuries to the legs
- 3. Rehabilitation after injury

According to information, the proportion of elderly people using crutches is more than 80%, it is evident that many elderly people are in need of devices to assist them in walking.

TARGET PRODUCT:

Combining the advantages and disadvantages of exoskeletons and senior crutches and correcting the product should be: Inconspicuous, Easy to wear, Not easy to lose.

General direction: The product should be a wearable joint support structure and the aid should be effective in helping the elderly to walk. And it should come with an adjustable module so that it can be worn by most people.



UCOND CRUTCHES

- Aluminum alloy material.
- Highlighted by lightweight + reinforced support.
- Combined with a complete arc bracket surface, it has a good decompression and anti-shock effect.
- It is also comfortable to walk for a long time, and the extra reinforced aluminum alloy tube is 1.2mm thick, which will be stronger and more stable to use.



GOOD WALKER WALKING STICK

- The selling point is the freedom of expansion and contraction.
- At the same time, the height can be adjusted at will, and the use of it is stable and also avoids slipping.
- It is made for people who have difficulty in walking and can bear weight up to 90kg.



LELAIFE CRUTCHES

- Keep the same with the previous model can be retracted.
- The selling point is that the night walk is also specialized with lighting function, not afraid of no place to go.
- And the base can be single can also switch 4 feet, through the universal rotation experience will be better.



FISH LEAP CANE

- This is a lightweight mobility aid.
- There are 9 adjustable levels of authority, including 5 levels in the grip area, and the feel is comfortable and non-slip.



INARI CRUTCHES

- It is made of stainless steel.
- Support multi-step adjustable, the main pipe used has the advantages of strong and reliable, and strong stability.



MEIDSTER

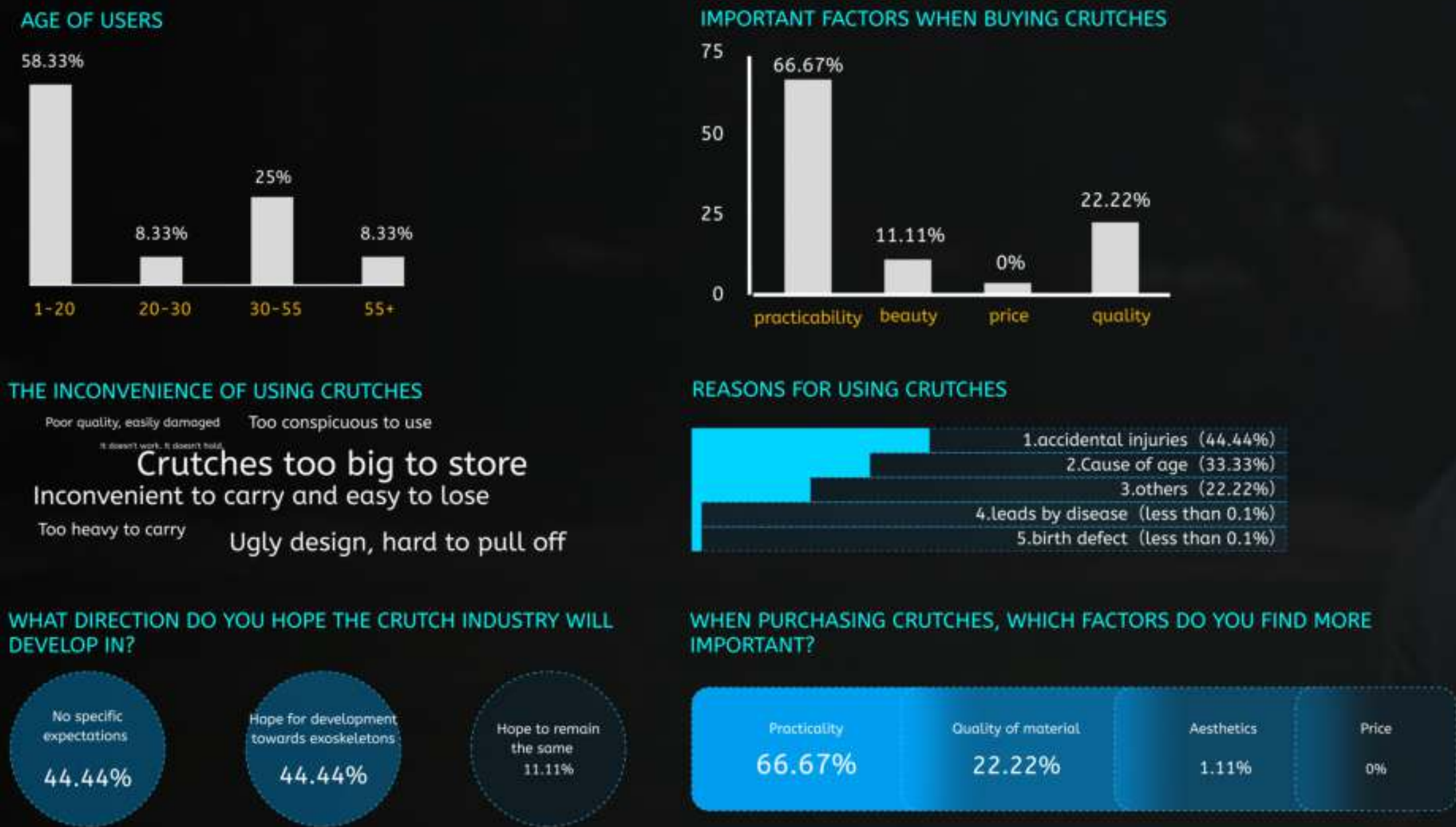
- Aluminum alloy.
- Often used in pulley-powered walking.
- It can be folded up easily without splitting the extra position.

CONCLUSION

According to background research, there are many types of crutches on the market, but all of them have a series of common problems: such as not easy to carry, easy to lose, unstable and so on. There are many well-established brands at the moment, and the advantages of these brands can be combined in the subsequent design.

QUESTIONNAIRE

To gain a more detailed understanding of people's views on modern canes, I created a survey and distributed it to individuals of different age groups to collect their opinions on canes and their suggestions for improvements.



USER NEEDS

- Be developed and specialization**
Allow exoskeletons and supporting structures to be developed for specialized torso segment specialization.
- Be portable**
These walking aids need to be portable.
- Be replaceable**
Not all trunk functions can be replaced by exoskeletons or other walking aids, which can lead to muscle degeneration.
- Be Smart**
The exoskeleton can be integrated with smart modules to make the device more convenient to use.
- Be easily stored**
They should be as small as possible so that they can be easily stored.
- Convenient**
6. take up as little hand space as possible, e.g. no need to hold.

INTERVIEW



Jane Alice

17 | Female | High school student with a previous ankle injury.

PASSIONATE INDEPENDENT

- Used underarm crutches for half a month.
Severe sprains of the ankle that prevent movement of the ankle.
- The prolonged use of crutches has led to swelling in my hands, a consequence of the extended pressure. Moreover, the crutches themselves are heavy and inconvenient, especially when trying to open doors. Both hands are occupied, making this task particularly challenging.
- I would like a cane that is easy to carry and does not have sharp edges on the handle.

Hugo Lee

19 | Male | Undergraduate student, studying sports rehabilitation.

PUNCTUAL WITTY

- Have some internship experience in rehabilitation institutions.
- Exoskeletons provide advantages for individuals with reduced functionality, offering increased portability compared to traditional assistive devices. The future of mobility aids may focus on enhancing and tailoring specific body parts.
- The future of crutches will undoubtedly focus more on assisting specific joints.



PERSONA

JACK

Age : 16
Gender: Male
Monthly income : ¥ - -
City : Beijing

A boy who hurt his leg when play basketball.

STORY

sprain his leg when playing basketball, After medical treatment,now is using crutch to assist walking. he has allready use the crutch for a week, according to doctor, he still need to use it for two more week.

FRUSTRATION

he dont like the now days curtch, Jack think it is havy and hard to storage. He also assume that the curtch he use is unstable.

WANTS & NEEDS

He wat a more reliable curtch.

- 1.more stable
- 2.more Easy to storage

My Best Friend



Andrew

Age : 53
Gender: Male
Monthly income: ¥15,000
City: Beijing

Have Leg disease.

STORY

by the reason of all days walking in young age, now andrew have lot of disease on leg. he can not walk very well, need crutch to assis walking.

FRUSTRATION

- When andrew don't have crutch, he can not do anything which need walking.
- The crutch is ugly, andrew think use those things is shame.

PAIN POINTS

- He wish the curtch can be more easy to storage.
- He think curtch need to improve on outlook.

My Grandfather

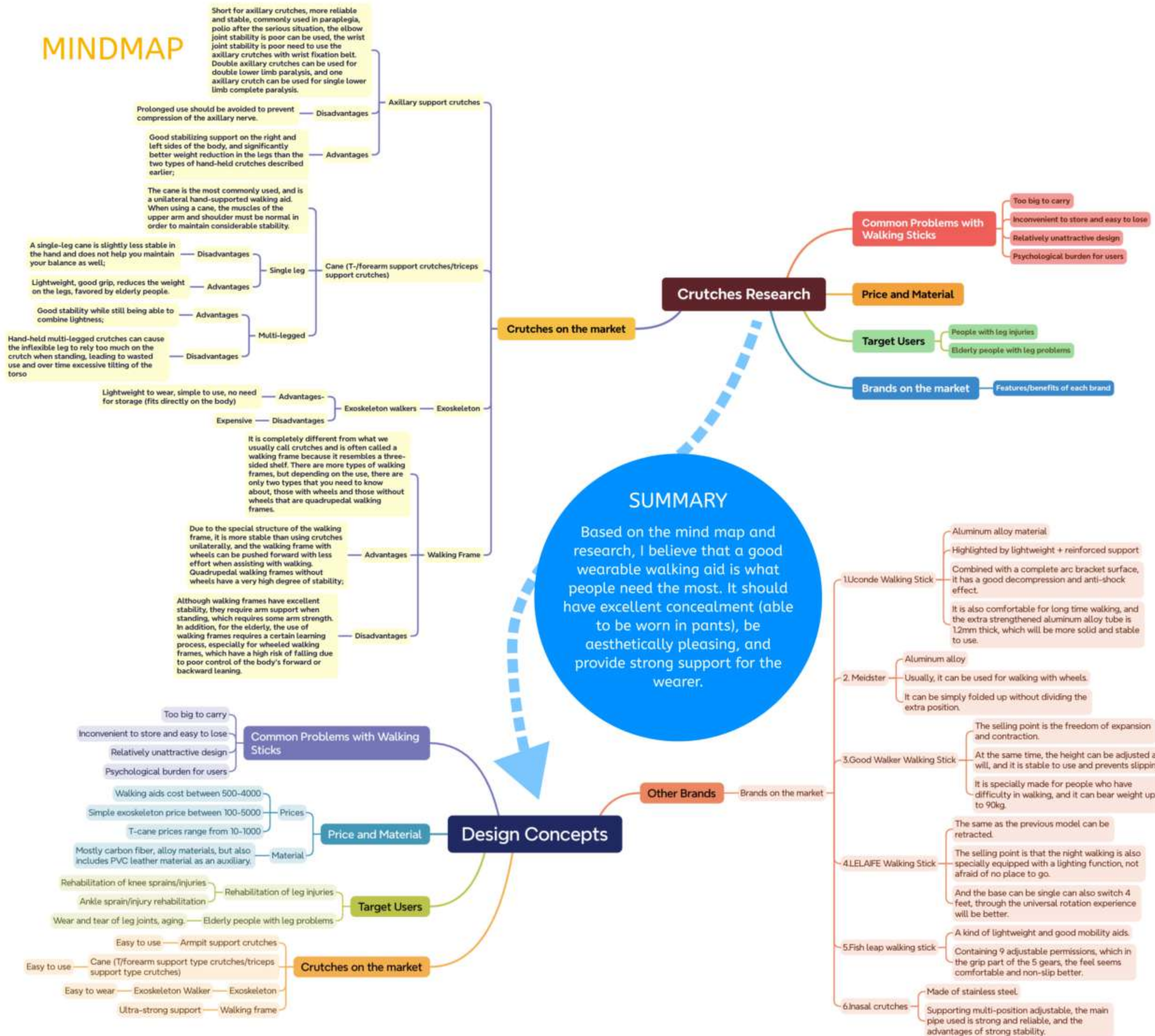


USER JOURNEY MAP

Jack Zhao 16 | Male | High school student, sports sprained leg, using support crutches, recovering.

STAGE	Getting up	on the way to school	Fetching crutches	Getting to school	Going to class	Eating in the cafeteria	Getting home from school	walking	Accidentally falling	Going home
USER BEHAVIOR	Getting out of bed and get his crutches.	he forgot his crutches.	Going home to get his crutches.	Going to class with his classmates.	Going to school	can't put crutches away	End of the day	It's a good thing he's got those crutches to hold him up.	Falling down in no man's land and no one noticing.	Go home and rest.
USER SENTIMENT										
ENVIRONMENT	Home to school (by car) at home	On the way home	On the way to the classroom school	on campus	On the way home from school on the way	on the way	at home			
USER PAIN POINTS	Crutches are sometimes placed far away No crutches.		There's no place for crutches near the desks.		The crutches are too big to put in the car				Unsteady crutches.	
OPPORTUNITY & CHALLENGES	Reduced crutch footprint.	Some reminder devices can be added or the crutches can be made to store well.	Make crutches good for storage.	Opportunity: make crutches good for storage. Challenge: easily stowed crutches are prone to damage.	Shrink the crutches or make them well stowed.			Opportunity: make the crutch's contact area with the ground larger, alarm function.		

MINDMAP



DESIGN CONCEPTS

ARMPIT SUPPORT CRUTCHES

ADVANTAGES

Good stabilizing support for the left and right sides of the body, and significantly better weight reduction in the legs than the two types of hand-held crutches mentioned earlier.

DISADVANTAGES

Prolonged use needs to be avoided to prevent entrapment of the axillary nerve.



CANES (T/FOREARM SUPPORT CRUTCHES/TRICEPS SUPPORT CRUTCHES)

ADVANTAGES

Lightweight, better grip, can reduce the weight of the legs, favored by elderly people.

DISADVANTAGES

Hand-held monopods are slightly less stable and don't help you balance as well.



EXOSKELETON WALKER

ADVANTAGES

Light to wear, simple to use, no need to store (directly on the body to carry), not easy to lose.

DISADVANTAGES

Expensive.



WALKING FRAME

ADVANTAGES

More stable than using crutches unilaterally, and the walking frame with wheels is propelled forward with less effort when assisting with walking.

DISADVANTAGES

Require arm support when standing, which requires a certain amount of arm strength. Higher risk of falling due to poor control of the body's forward or backward leaning.



Final Concept



Through comparison, I finally settled on an exoskeleton style walking aid.

The advantage of this device is that it is better able to assist with specific joint movements, while also making the assistive structure light enough.

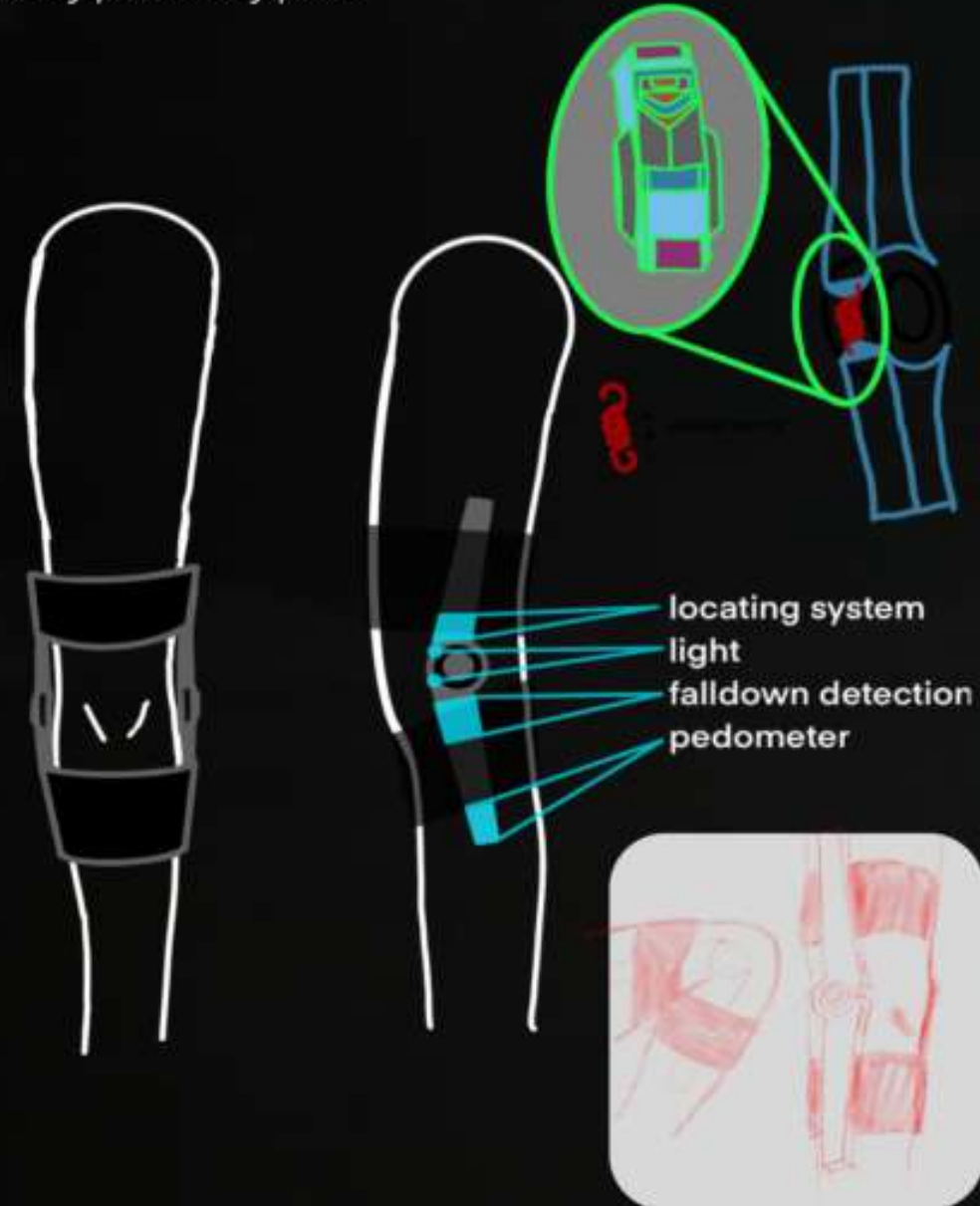
Unlike crutches, which require the upper body to be attached to the ground for support, the exoskeleton only needs to be wrapped around a specific injury to aid in walking.

FUNCTIONS THAT CAN BE ACHIEVED:

- Easy to wear and carry.
- Assisted knee and ankle support.
- Fall detection (start contacting family members after a long period of time in a lateral position after rapid shifting while wearing the exoskeleton).
- Connection to cell phone app to locate the device.

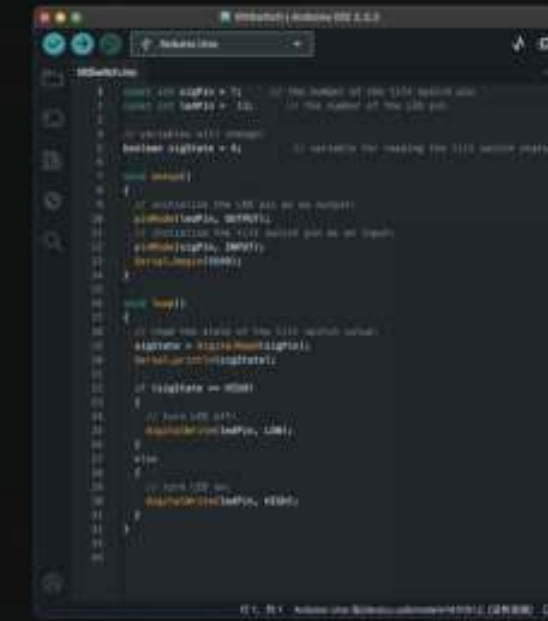
SKETCH

Based on the conclusions drawn earlier, I designed two different wearable assistive devices: one for full leg support and the other for knee support. The structure for full leg support is relatively stable and strong, but it is complex to wear and difficult to carry. In contrast, the knee assistive structure not only provides support but is also easy to carry. Therefore, I believe the knee assistive structure will be my preliminary plan.



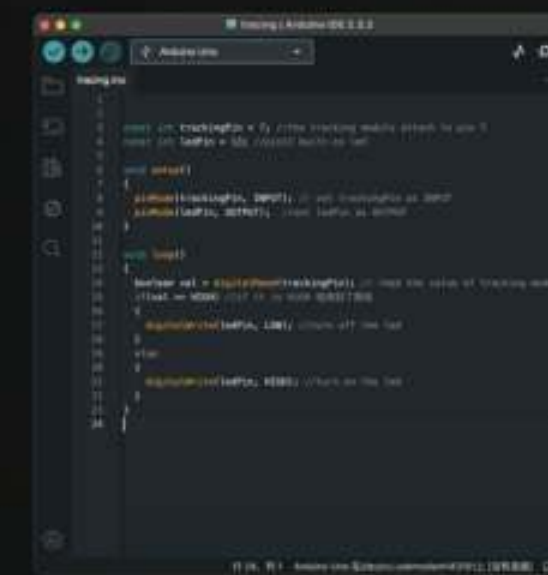
INTERACTION & CODING

The assistive device is equipped with a positioning function and a fall alert feature, allowing doctors and family members to arrive at the scene promptly in the event of an injury to the wearer.



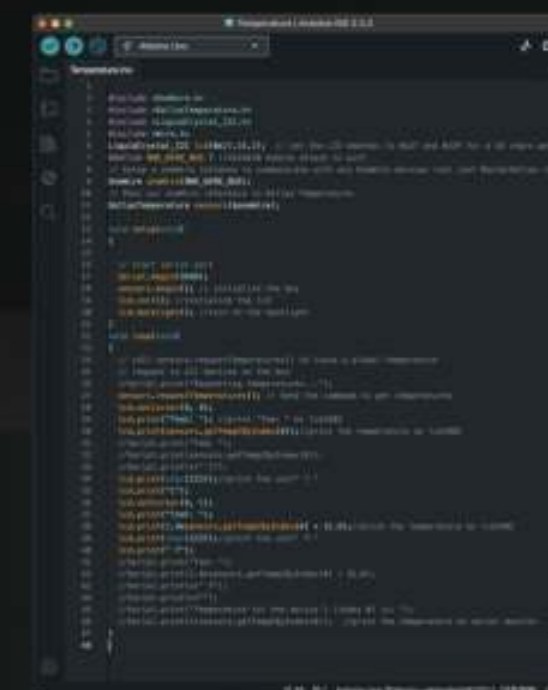
FALL MONITORING

The product is equipped with a gyroscope and vibration sensor. When it detects that the user has fallen and cannot save themselves, it will send location information to family members or doctors.



POSITIONING MONITORING

Built in GPS positioning device helps users' family or doctors quickly locate patients for rescue operations.



TEMPERATURE MONITORING

Built in lower limb joint temperature monitor, when the user experiences abnormal heating due to excessive exercise or other reasons, the cooling pad will start working to alleviate the user's discomfort symptoms.

3D MODEL & PRINTING



This is the final modeling part of my structure, which consists of two fixed rods and an axis. The fixed rods have elongated openings to facilitate the wearer's use of Velcro. The overall design features streamlined aesthetics for a more appealing look. Additionally, I've added a maximum angle limit to prevent the wearer's knee from bending at too great an angle.



USER TESTING



ASSUMPTION



PRODUCT



Walking assistance



Status monitoring



Information transmission



GPS Location



Cooling



SPACE



Diagnosis of illness



Psychological counseling



Equipment exercise



Hospitalization

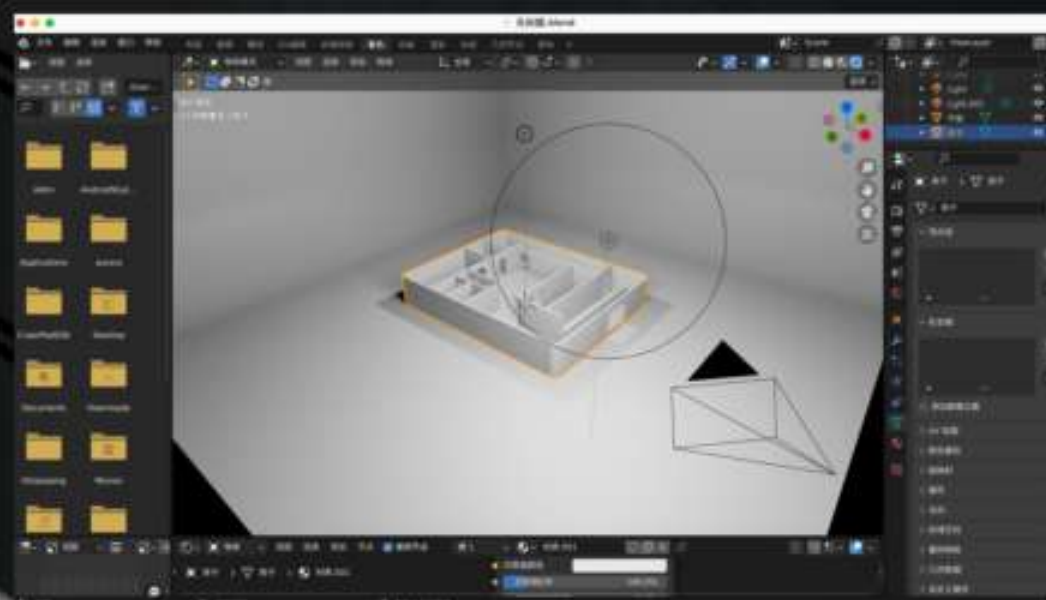


Patient communication

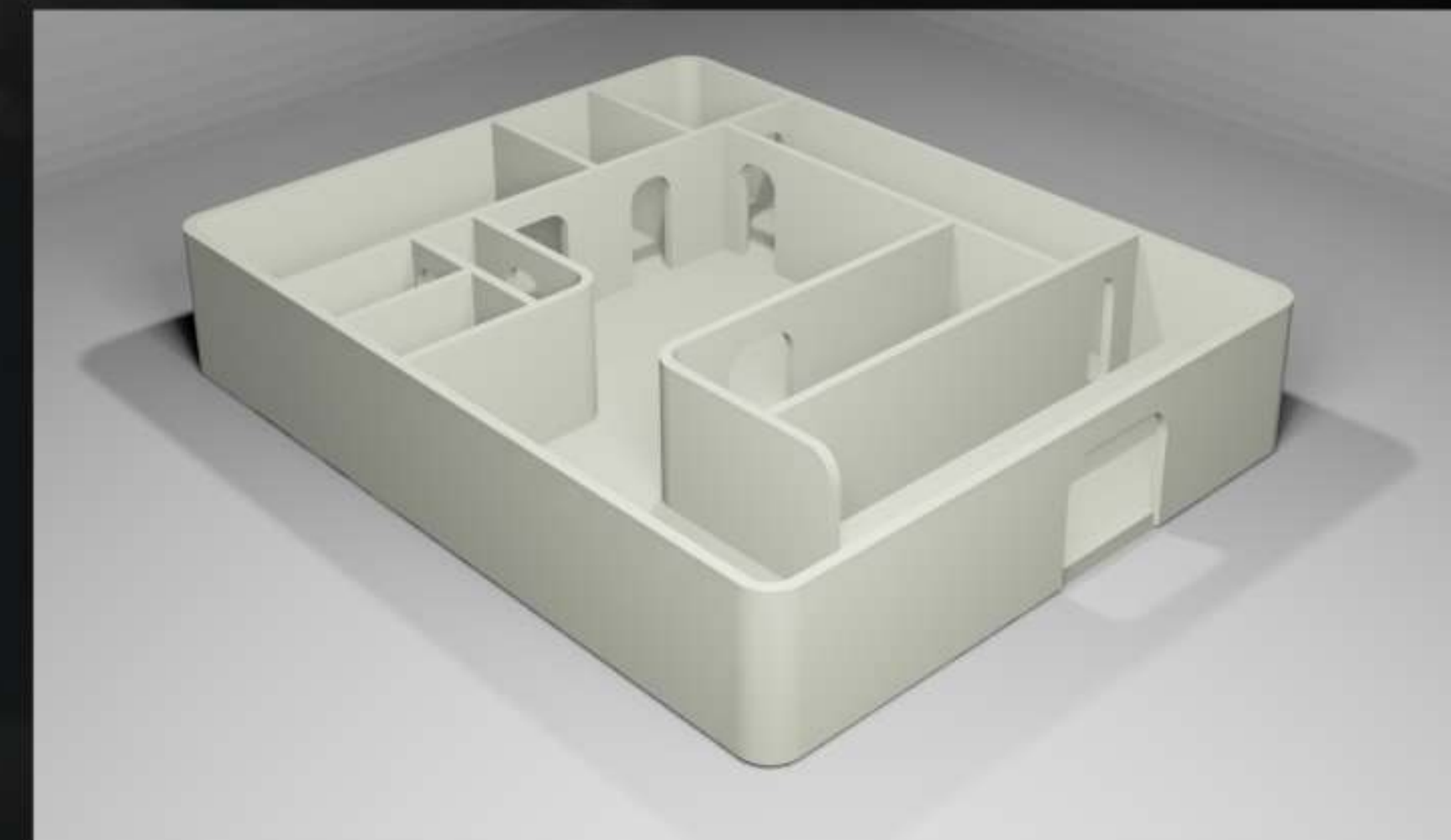
FILED INVESTIGATION



SKILL - BLENDER



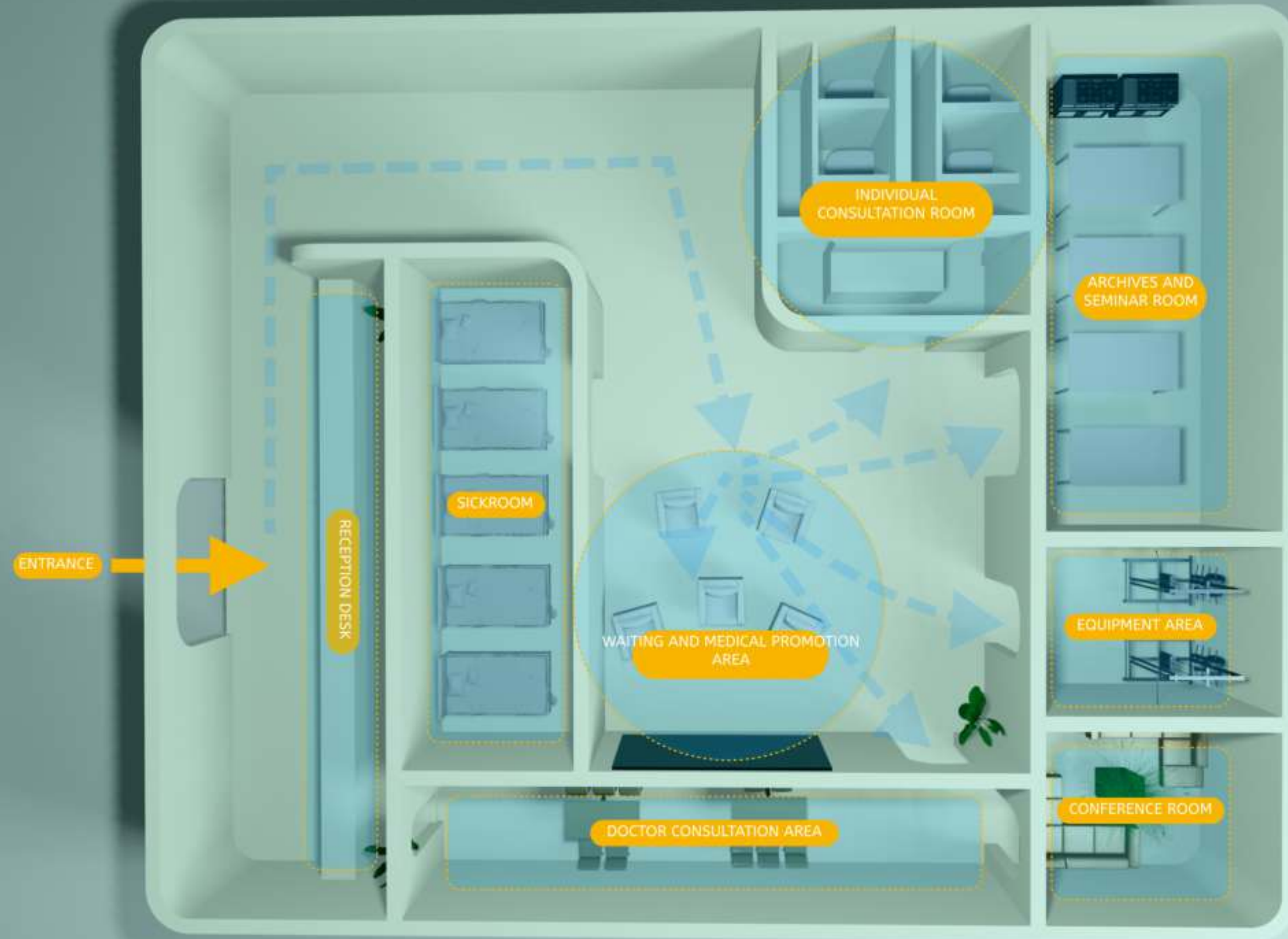
SERVICE SPACE RENDERING



FINAL WORK

- SERVICE SPACE DESIGN

In this project, my aim is to develop a novel assistive device akin to a cane. I have conducted research on the canes currently available in the market, gathering insights into their strengths and weaknesses. It is my aspiration that this project will effectively address these limitations and integrate as many advantages from different types of canes as feasible.

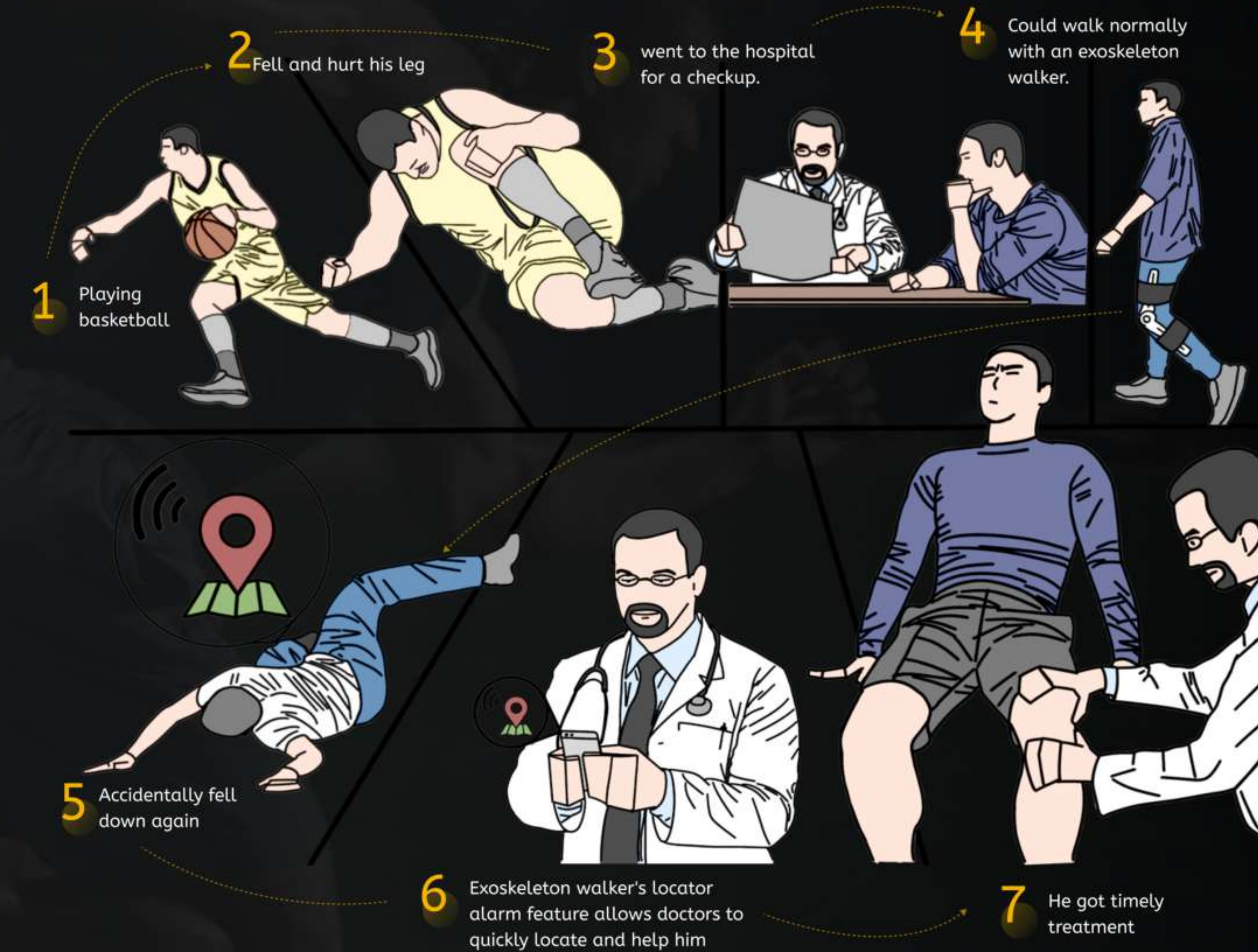


- PRODUCT DESIGN-EXOSKELETON WALKER

Exoskeletons walker help patients relieve the stress on their limbs, aid in their recovery, and transmit relevant information to the clinic by monitoring the patient's condition.



STORY BOARD



EVALUATION

TARGET USERS:

- Designed for all age groups.
- Primarily for rehabilitation training of patients with knee injuries.
- Also suitable for hikers and trekkers.

PRODUCT ADVANTAGES:

- Addresses the difficulties of using and carrying traditional canes.
- Provides better support and convenience.

FUTURE IMPROVEMENT SUGGESTIONS:

- Optimize the product's app.
- Add more interactive features.
- Enhance the product's intelligence.

INDUSTRIAL PRODUCT DESIGN

PROJECT 02

INTERACTION DESIGN

2024

INDIVIDUAL WORK

CHRONO SCENERY

PRODUCT & INTERACTION DESIGN

▶ <https://youtu.be/6owS25Ky1ml>

I designed an aesthetically pleasing task reminder to help students and professionals better manage their daily tasks. Often, existing reminders are dull or stress-inducing due to their serious design. My task reminder is both beautiful and practical, aiming to reduce stress and prevent overlooked tasks.

KEYWORDS *Procrastination, Reminder, Scenery, Interesting, Interactive product*

MATERIAL *Arduino board, Toggle switch, Smart servo, Acrylic sheet, Laser cutting machine*

SIZE *Product - Chrono Scenery 98cm*58cm*9cm*



BACKGROUND RESEARCH



In order to better understand what people do throughout the day and what kind of assistance they need to ensure they can complete tasks on time, I conducted some research and found some conclusions.

STUDENT: 6 AM TO 11 PM (17 HOURS IN TOTAL)

Besides attending classes (about 8 hours), school homework is about 2-3 tasks (about 1.5 hours), extracurricular assignments are about 1-2 tasks (about 1 hour), entertainment (gaming) and leisure (napping, walking, exercising), eating (three meals a day) will take up the remaining time.



WORKER: 8 AM TO 12 AM (15 HOURS IN TOTAL)

Besides working hours (about 8 hours), part-time jobs (social media, night shift customer service) (about 3 hours), tidying up the house (about half an hour), entertainment (gaming) and leisure (napping, walking, exercising), eating (three meals a day) will take up the remaining time.



WAYS OF NOTING DOWN TASKS:



STICKY NOTES

Advantages:
Very convenient, clear for note-taking, can write anything on them.
Disadvantages:
Easy to lose, sometimes forgotten to be checked.



USING YOUR MEMORY

Advantages:
Quick and effective, easy to remember.
Disadvantages:
Easy to forget.



ALARM

Advantages:
Timely and effective, phone alarms can also have text reminders.
Disadvantages:
Can disturb others in public places, can be unpleasant when waking up during rest.



CALENDAR

Advantages:
Fixed location, hard to lose, effective reminders.
Disadvantages:
Reminder time is not precise, usually only reminds the date, can't fit many tasks, and cannot remind specific times.



NOTEBOOK

Advantages:
Can clearly record information in an organized manner.
Disadvantages:
If lost, all records are lost. Sometimes forgetting to check the notebook can lead to missed tasks.



SMARTWATCH

Advantages:
Portable, timely and quick reminders.
Disadvantages:
Useless if forgotten.



NOTES/MEMO APP

Advantages:
Quick and effective, can record directly on electronic devices, can include pictures.
Disadvantages:
Lacks reminders, sometimes can be forgotten, may miss



LIST

Advantages:
Clear organization, quick reminders.
Disadvantages:
Recording can be complicated and tedious.

MARKET RESEARCH



SCHEDULE MANAGER SCHEDULE ALARM, VISUAL CHART



Advantages:

Simple and quick, clear information display, progress visualization.

Disadvantages:

Needs to be used on electronic devices, no reminders if the phone is not nearby.



DESKTOP STICKY NOTES SCHEDULE PLANNING CHECKLIST, PUNCH CARD MEMO POST-IT



Advantages:

Simple and aesthetically pleasing, available as soon as you open the computer.

Disadvantages:

Must open the computer to know the to-do items, takes up additional running space.



TO-DO LIST NOTEBOOK

Advantages:

Can be recorded manually, visually effective.

Disadvantages:

If you forget to check the notebook, it won't serve as a reminder.



MESSAGE BOARD

A WHITEBOARD, WITH MAGNETS THAT CAN STICK TO THE COMPUTER.

Advantages:

Convenient to use, can be reused multiple times.

Disadvantages:

Easily scratched accidentally, inconvenient to carry.

CONCLUSION



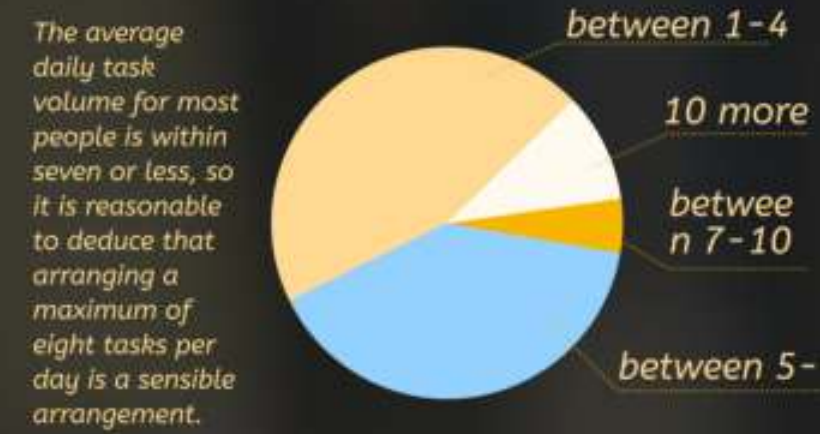
- 1 Most note-taking products have drawbacks such as being easy to forget or easy to lose.
- 2 Many note-taking products are not integrated with clocks, making it easy to forget to-do items.
- 3 All note-taking products require a custom event module to allow users to fill in their own tasks.
- 4 Note-taking products combined with clocks or progress bars are often more intuitive and user-friendly.

USER RESEARCH

Questionnaire

I sent out the questionnaire to family members, friends, and teachers, and received a total of 39 samples from different age groups. Based on these samples, I have drawn the following conclusions:

User Daily Task Volume Survey



The Frequency of People Forgetting Tasks



The main reasons for not completing tasks on time

Forgetting
Anxiety
Not being reminded
Not wanting to do it
Having too many things to do
Being playful
Escaping

Interview

To gain a more nuanced understanding of the issue, I interviewed three classmates.



Kevin | high school student

I have 6-8 tasks to do every day, and most of the time I can remember them, but sometimes I forget to do them. Occasionally, I use the phone's memo to record tasks, but when I'm busy, I might also forget to check the memo.

Aubert | high school student

I have about 4 tasks to do every day and I like to use the memo to record tasks. I rarely forget to do them. However, sometimes I might miss a task because the memo doesn't remind me.



Kassy | high school student

I usually complete my daily tasks, but sometimes I forget to do them. I prefer to use the phone's memo to record tasks, but occasionally the phone distracts me, making it hard for me to focus on my tasks.

TASK USERS

Workers

Due to the lack of suitable time management tools, they always procrastinate on completing projects. They are not good at arranging their time. They need some devices to help them plan their time.



Students

Due to the lack of good reminder tools, the assignments that need to be done are always forgotten.

They need a conspicuous device to remind them of the tasks they should do.



USER NEEDS

- 1 Reminder for Pending Tasks.
- 2 Time Alerts.
- 3 Non-intrusive.
- 4 Assistance in Time Planning.
- 5 Strong Reminder Capability.



Jack Morrison

Age: 29 years old
Education: Bachelor's Degree
Job: Office Worker

SCENARIO

Jack's job is demanding, with frequent deadlines and meetings. He often doesn't know what to prioritize, so he frequently forgets to do some work.

GOALS & EXPECTATIONS

Efficient Time Management: Jack wants a tool that can help him manage his busy schedule and prioritize tasks.
Task Prioritization: He needs a way to view and organize his tasks based on importance and deadlines.
Reminders and Alerts: Jack expects the app to send reminders for upcoming deadlines and meetings.
Usability: He prefers an app that is user-friendly and available on multiple devices.

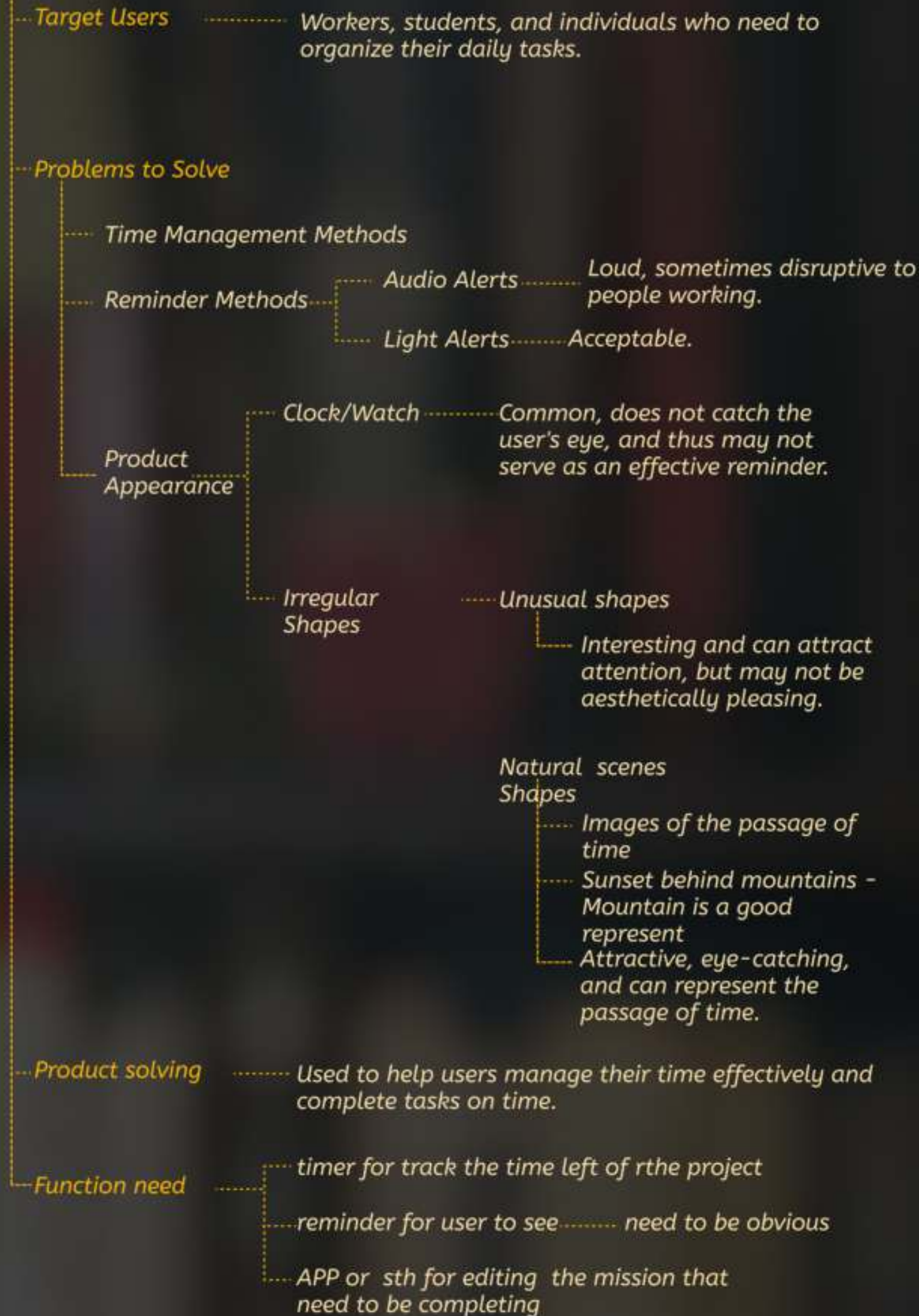
USER JOURNEY MAP

Time in day	0:00-8:00	8:00-8:30	8:30-11:30	11:30-13:00	13:00-17:00	17:00-20:00	20:00-20:30	20:30-21:30	21:30-24:00
Stage	sleeping	breakfast	working	lunch break	working	part-time job	dinner	house work	playing game / relaxing
Behavior	relax	start to schedule the day	have lot work to do, and may confuse about where to start	break	have lot work to do	have lot work to do	break	have lot work to do, and may confuse about where to start	break
Sentiment									
location	bedroom	canteen	company	canteen	company	work area	canteen	home	home
Pain points		The user may not have a really intense sense of time, which could lead them to being late for wrok.	Users may lack a well-structured time plan, leading to uncertainty about where to begin.	Users might not have a clear time management strategy, which can cause a lack of direction and disrupt their planning.	He often feels overwhelmed and doesn't know where to start, leading to a disorganized approach to his tasks. For example, instead of setting a morning goal to finish a report, he might dally with less urgent matters, causing important work to pile up and resulting in a chaotic day with unfinished tasks.		He may not have a coherent schedule, which can lead to a weak perception of time management. This often results in difficulties for him in determining the initial steps necessary to proceed with his tasks, creating a sense of uncertainty and potentially causing disarray in his overall planning.		
Opportunities		Time reminder	A device can assist in organizing tasks that need to be done or prompt users to schedule their daily activities, subsequently reminding them of their tasks based on the arranged times.		Time reminder	A gadget can help to manage tasks that are pending or guide users in planning their daily routines, then alerting them of their duties according to the set timings.		A device can facilitate the organization of pending tasks or aid users in arranging their daily schedules, subsequently notifying them of their assignments at the designated times.	

INSPIRATION

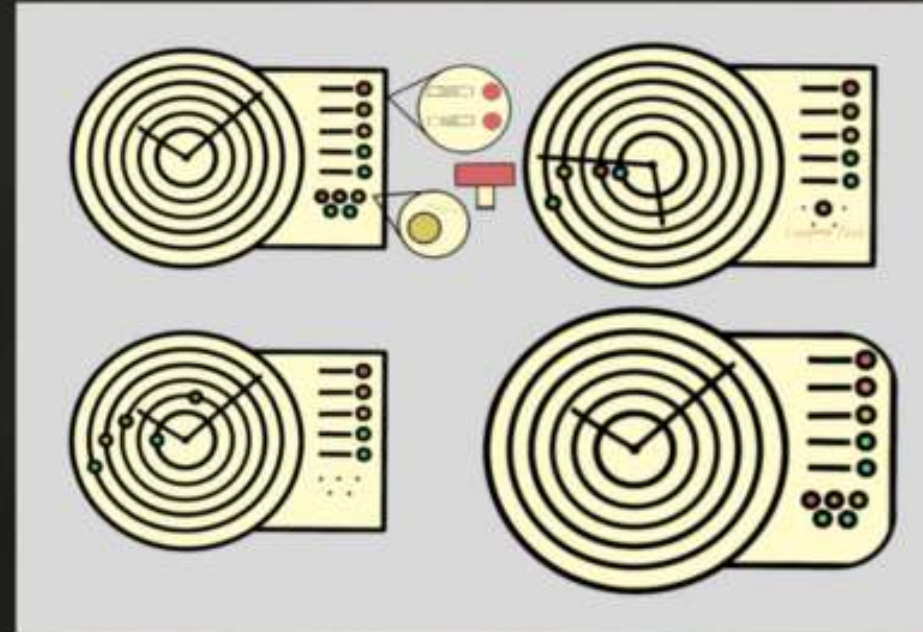
Based on the survey questionnaires and interviews, I believe that people are lacking a device that can remind them to do tasks in a timely manner and help them plan their tasks. With this concept in mind, I have made the following rough sketch designs.

Design Considerations



DESIGN CONCEPTS

Concept 01



Tack clock

Place tacks as task reminders on a dial with pointers that are pushed by the pointers as time passes.

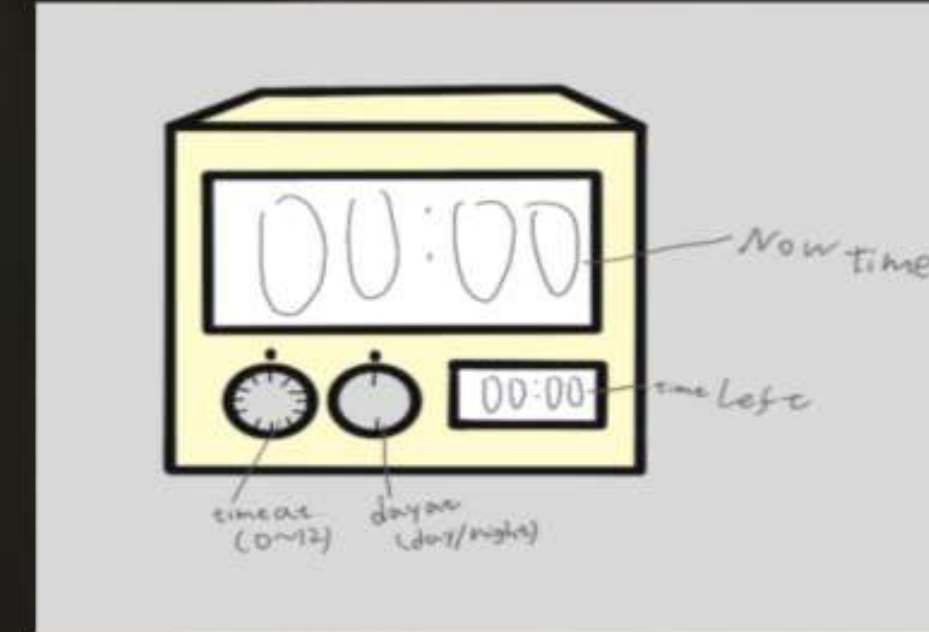
Pros

- Simplicity
- Easy to use
- Can edit to-do list

Cons

- Low maneuverability

Concept 02



Event alarm clock

A large screen displays the present time, a small screen shows the time remaining for the task, and two knobs are used to set the time for the task.

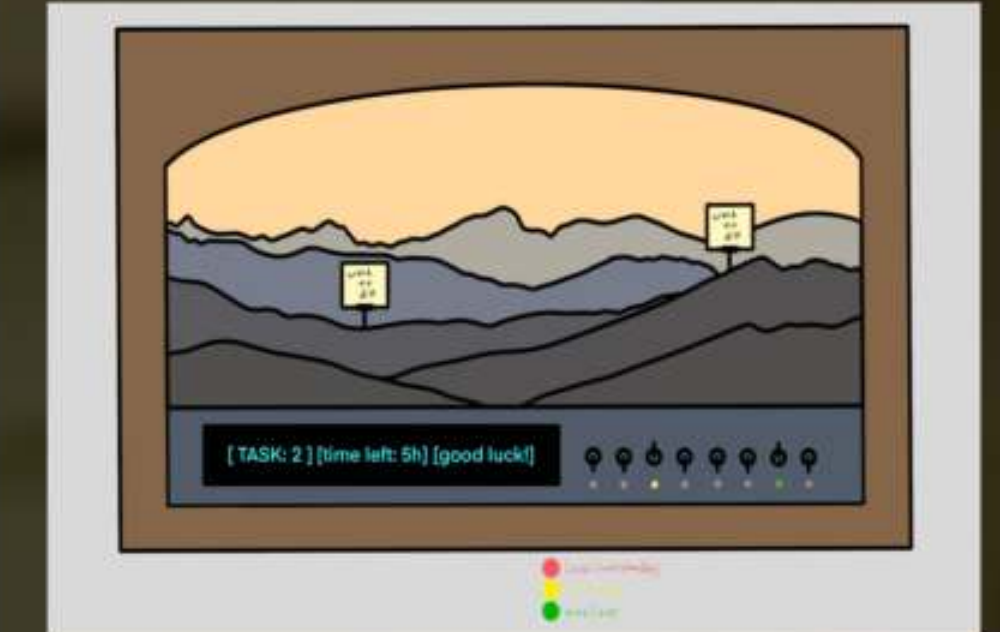
Pros

- Easy to use
- clear goals
- interesting shape

Cons

- Can only schedule one task

Final Concept



ChronoScenery

The mountain gradually darkens from left to right to represent the passing of the day, and the sticky notes on the mountain represent the tasks that need to be done at a specific time.

Pros

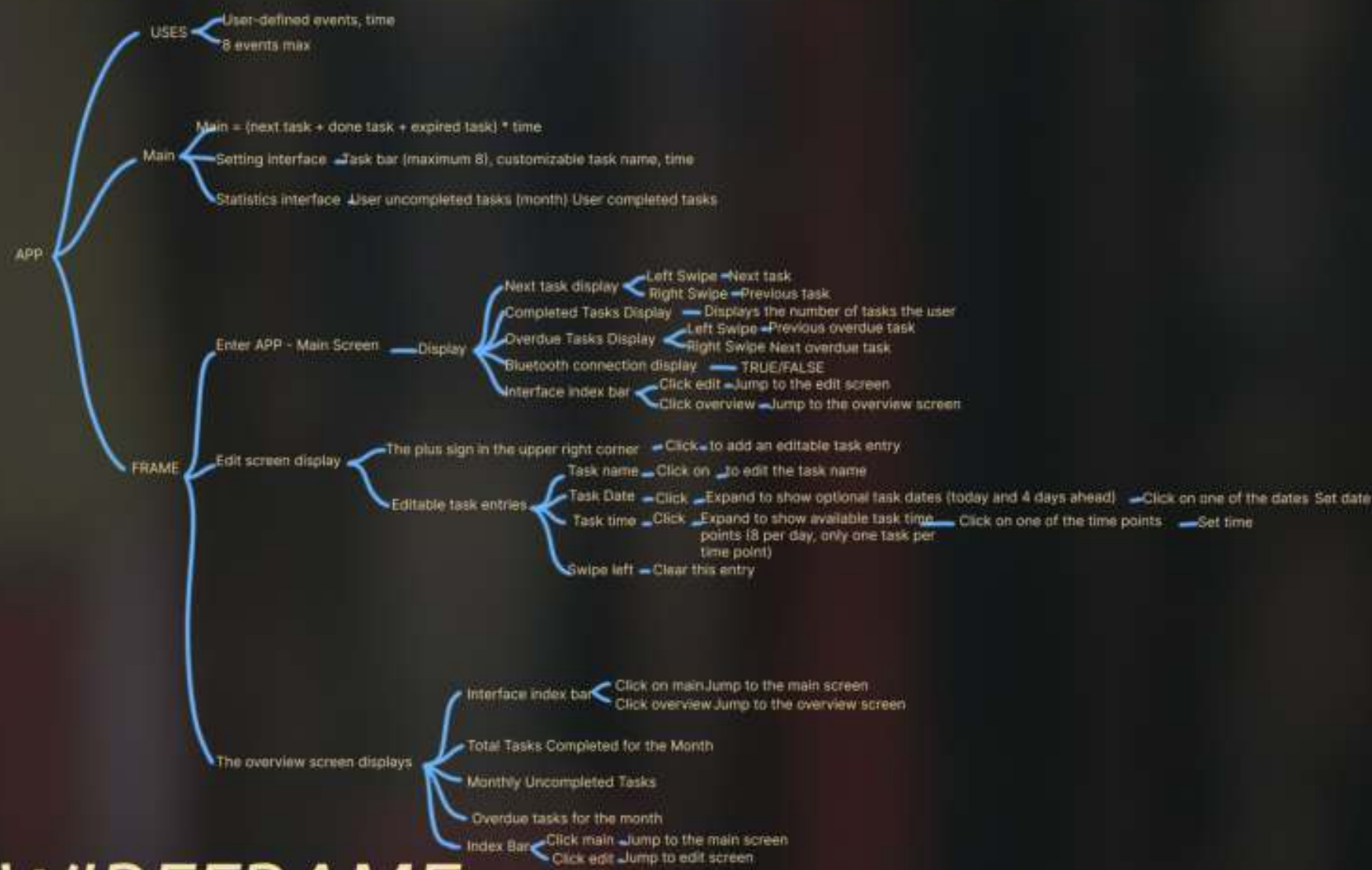
- Simple to use
- Can edit to-do list
- Interesting shape
- Clearly defined goals

Cons

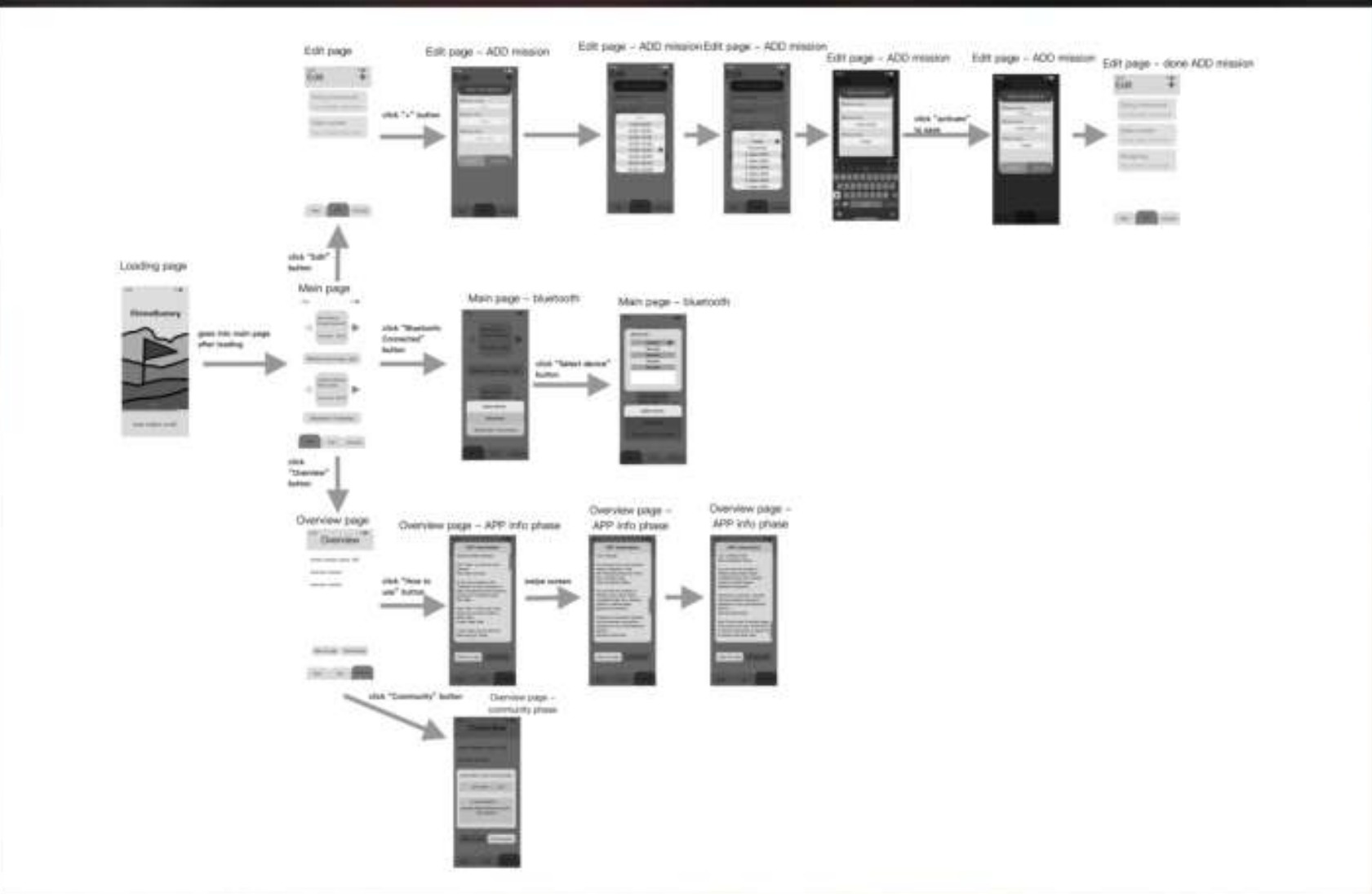
- Larger than other designs

STRUCTURE FLOW

I developed an APP that assists users in customizing task bars, editing tasks, conducting monthly statistics, and connecting to a Bluetooth calendar, all to help users manage and plan their daily tasks more efficiently.



WIREFRAME



MAIN INTERFACE

Loading Screen
The first screen displayed when entering the software.

Home Screen
Provides an overview of the task completion progress and informs the user of the tasks that still need to be completed.

Overview Screen
Records some basic information about the user and the software.

Edit Screen
Used to overview all the tasks set by the user.

Add Task
By clicking the plus sign on the edit screen, you can add a new task.

User Manual
Displays the method of use for all the software's functions.

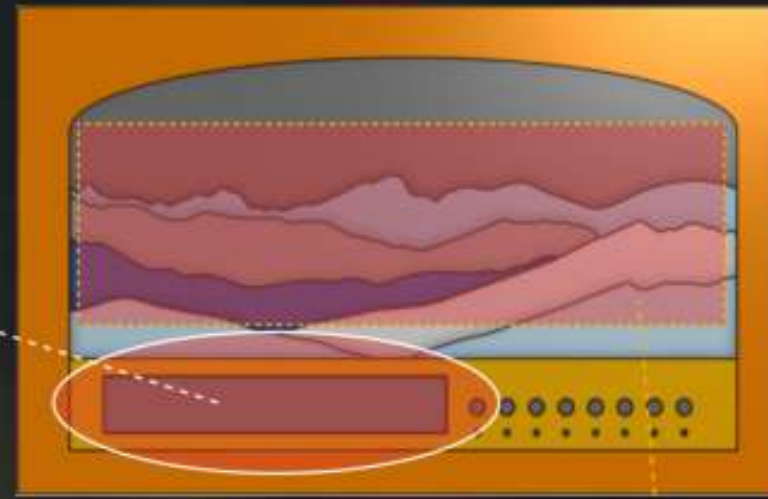
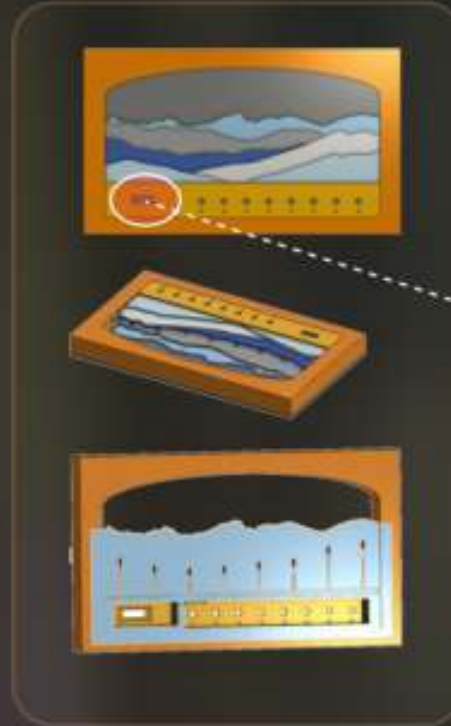
Bluetooth Connection
This section records the devices the user has previously connected to.

PRODUCT DESIGN

Product 3D modeling

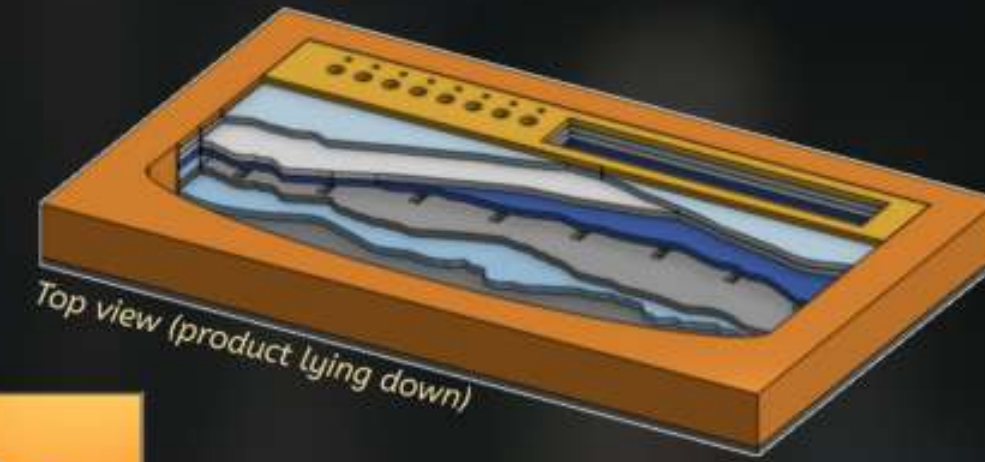
Small screen version

Large screen version



Front view

The relative position of the sun and the mountains was often the best way for people to determine the time in the past. My calendar app also uses this design, with the light bar at the top simulating the sun illuminating the mountain peaks, which darkens as time passes.

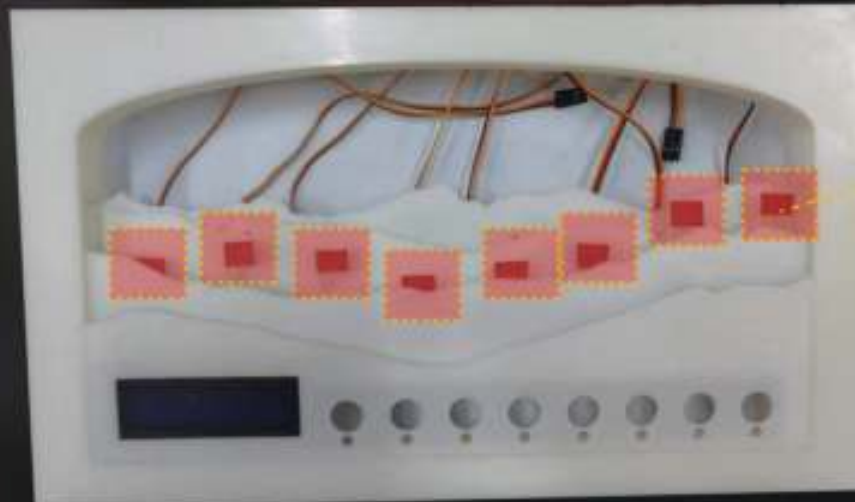


Top view (product lying down)



Back view

Photosensitive resin laser printing



Additionally, flags are a commonly used form of marking, and they are used because flags are most suitable for appearing in mountainous terrain.

Installation of circuits and sensors

Installation of printing materials



CODING

```
#include <Servo.h>
#include <FastLED.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <NewPing.h>
```

```
Servo a;
Servo b;
const int buttonPin = 2;
const int buttonPin2 = 7;
const int ledPin = 7;
```

```
const int ledPin2 = 12;
int lastButtonState1 = 1;
int lastButtonState2 = 1;
bool one = false;
bool two = false;
int buttonState = 0;
int redPin = 3;
int greenPin = 5;
int bluePin = 6;
int buttonState2 = 0;
void setup() {
  Serial.begin(115200);
  pinMode(ledPin, OUTPUT);
  pinMode(ledPin2, OUTPUT);
```

```
pinMode(buttonPin, INPUT);
pinMode(buttonPin2, INPUT);
a.attach(9);
b.attach(10);
Serial.println("C");
pinMode(redPin, OUTPUT);
pinMode(greenPin, OUTPUT);
pinMode(bluePin, OUTPUT);
}
```

```
void loop() {
  delay(1000);
  buttonState = digitalRead(buttonPin);
  buttonState2 = digitalRead(buttonPin2);
```

```
if (buttonState !=
HIGH&&lastButtonState1!=buttonState) {
  one=!one;
}
lastButtonState1 = buttonState;
if (one) {
  digitalWrite(ledPin2, HIGH);
  a.write(180);
  Serial.println("A");
}
else {
  digitalWrite(ledPin2, LOW);
  a.write(0);
  Serial.println("B");
}
```

```
if (buttonState2 !=
HIGH&&lastButtonState2!=buttonState2) {
  two=!two;
}
lastButtonState2 = buttonState2;
if (two) {
  digitalWrite(ledPin, HIGH);
  b.write(180);
  Serial.println("A");
}
else {
  digitalWrite(ledPin, LOW);
  b.write(0);
  Serial.println("B");
}
```

```
void colorRGB(int red, int green, int blue){
  analogWrite(redPin,constrain(red,0,255));
  analogWrite(greenPin,constrain(green,0,255));
  analogWrite(bluePin,constrain(blue,0,255));
}
```

Sensor interaction instructions

This machine utilizes a variety of components, including:



1. Servo Motors: Located beneath each chess piece, they are linked to buttons and controlled by the APP to raise or lower the flags.



2. Buttons: Positioned at the bottom right of the device, they are pressed by the user when a task is completed, which then controls the servo motor to lower the flag.



3. Display Screen: Located at the bottom left of the device, it shows current tasks, remaining time, current time, and other information.

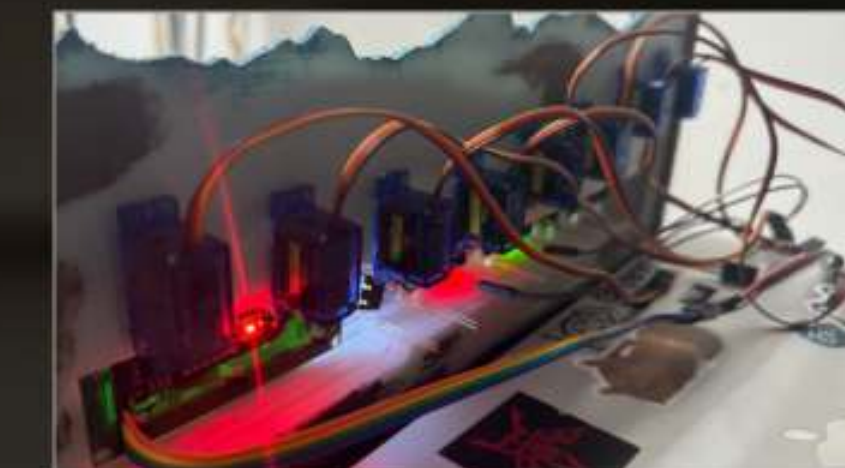


4. Light Bar: Situated on the inner upper half of the device's frame, it dims from left to right over time and resets at the start of a new day.



5. Arduino Mainboard: Responsible for controlling the overall logic and feedback.

Arduino component working test



FINAL WORK

Product Status 1



One red, one green Display screen shows the next task The user has set two tasks, and the display screen prioritizes showing pending items, with overdue items changing from a green light to a red light.

Product Status 2



One red Display screen shows overdue tasks A task set by the user is overdue, while all others have been completed.

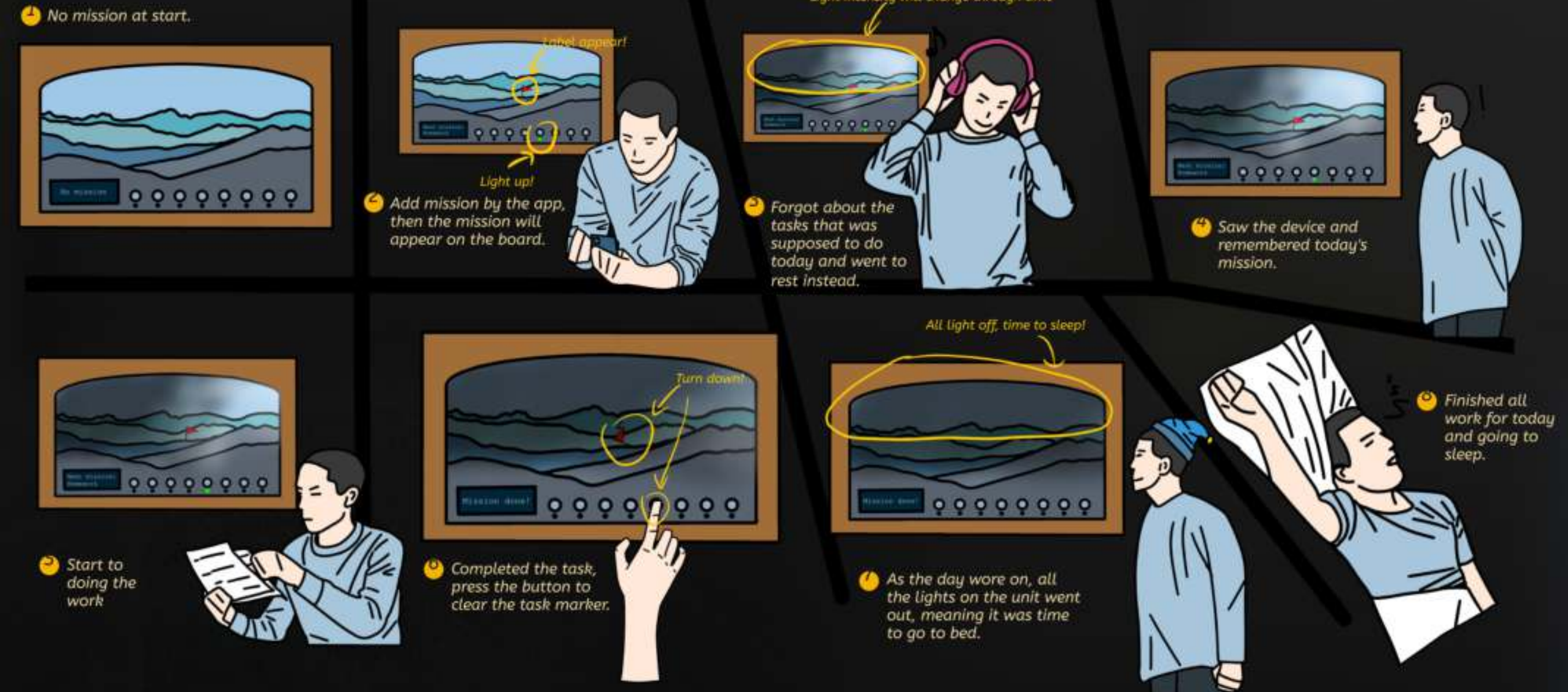
Product Status 3



None Display screen shows all tasks completed Once all tasks are completed, it returns to the initial state.



STORYBOARD



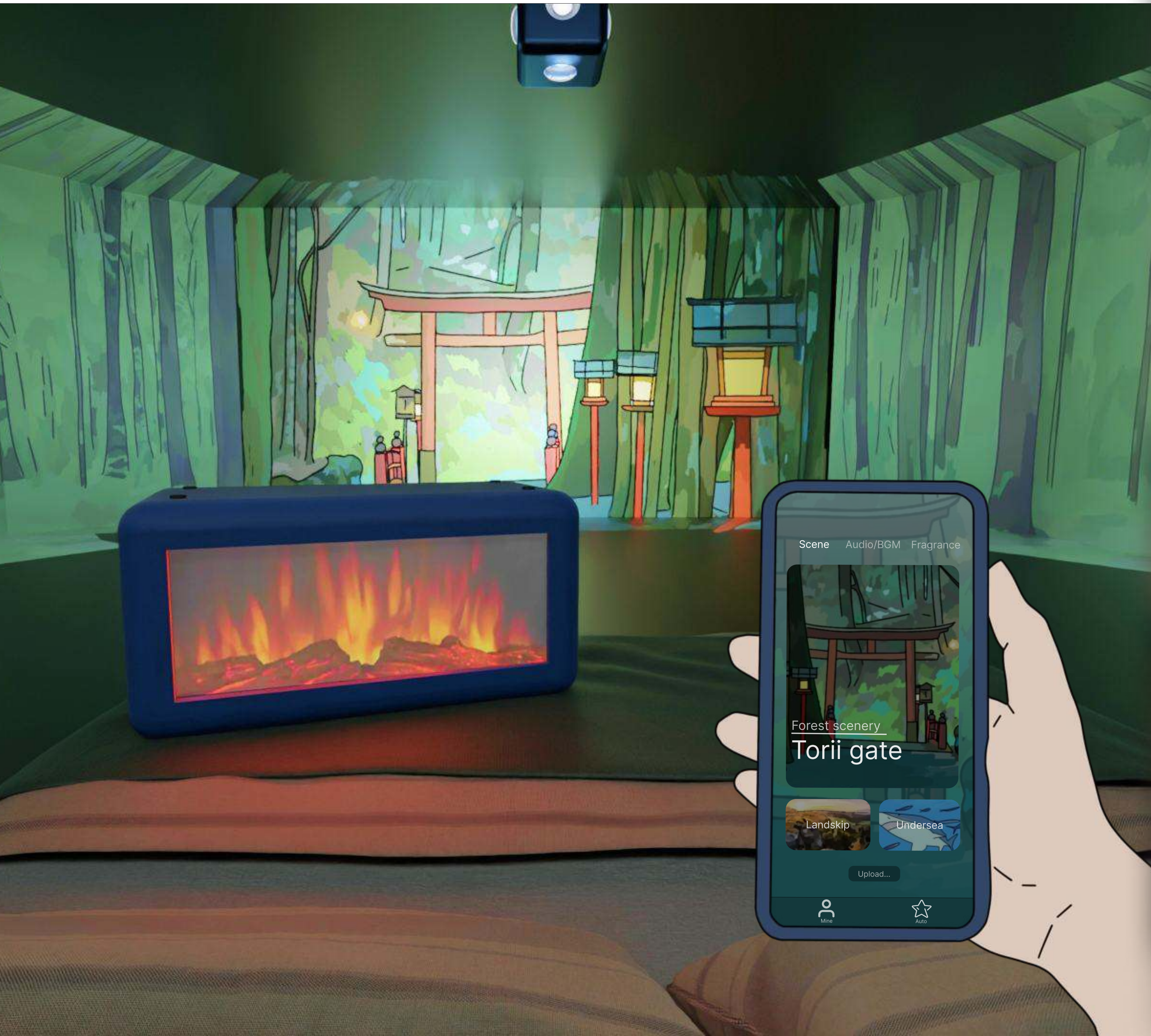
<https://youtu.be/6owS25Ky1ml>

SUMMARY



I believe this project can successfully help people better plan their time and complete tasks. However, there are still areas that can be improved:

1. There could be stronger interaction, perhaps requiring some audio reminders.
2. Maybe all users' devices could be interconnected so that they can see others' task progress.

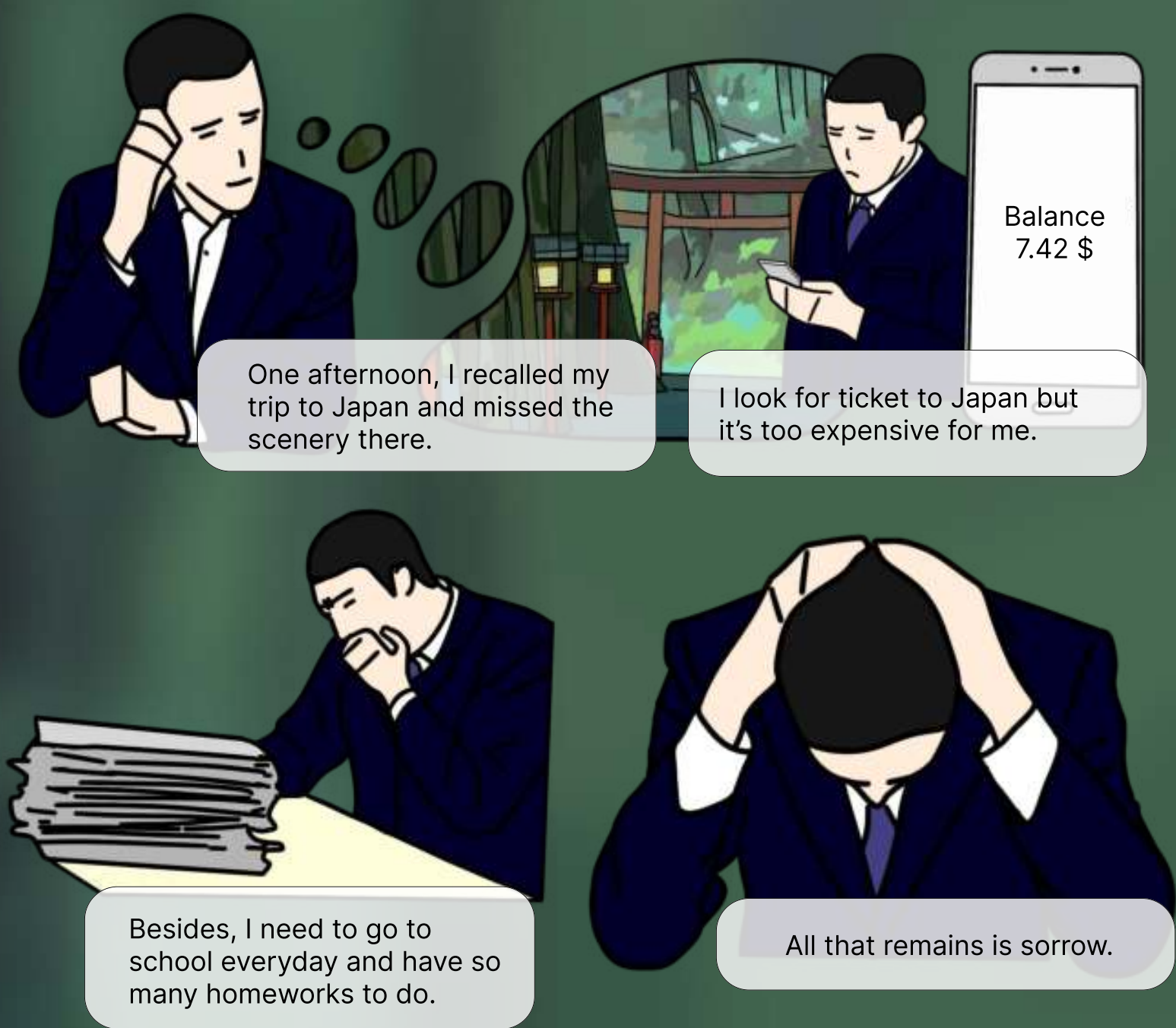


OmniTraveler

Interactive design
Product design
Experience design

I designed a device that lets people enjoy scenic views from home, aiding those with time constraints, physical limitations, disabilities, or in palliative care. In today's fast-paced society, many find it difficult to travel due to career or study commitments. This innovative product allows busy individuals, including those with special needs, to experience distant landscapes from home or during breaks, helping them relax and relieve stress.

Inspiration

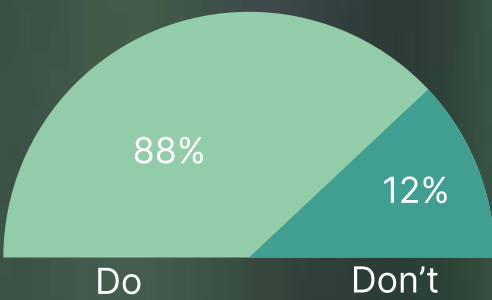


Background

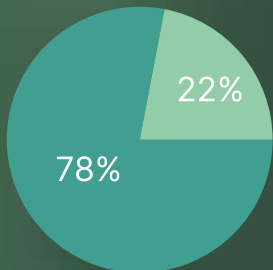
Recently, many people around me have been traveling less and less. Some are too busy with work, others are overwhelmed with their studies, and some lack the financial means. I want to find a way to solve or at least alleviate the issue of people not being able to travel. But before that, I need to understand how widespread this phenomenon really is.

Questionnaire

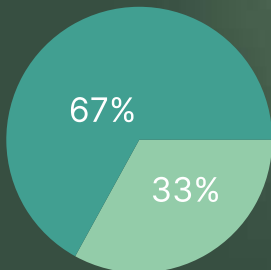
People's Willingness to Travel



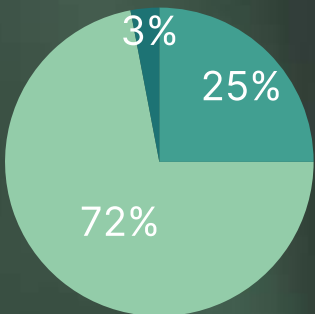
Do people want to visit beautiful scenic spots again?



Workday Travel Preferences:



Holiday Travel Preferences:



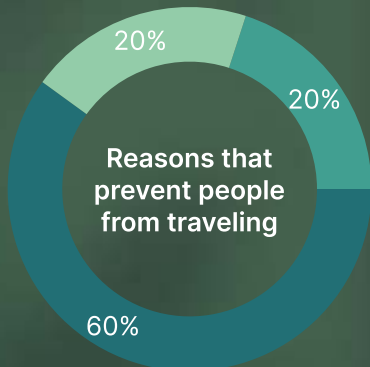
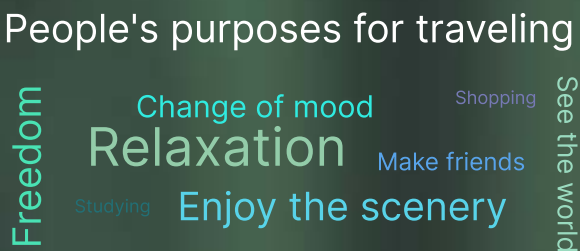
Like
Natural
Dislike

People's preferences for tourism

Conclusion

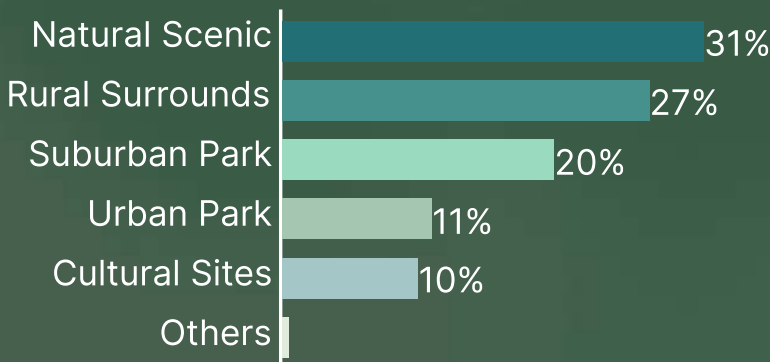
I found that people really enjoy traveling, are willing to go sightseeing, and cherish the beautiful scenes they've seen before.

Purpose of travel:



Time reasons (too many things to do)
Financial reasons (not enough money)
Energy reasons (too lazy to move)

People's Preferences for Travel Destinations



Conclusion

After conducting research, I found that this issue is not limited to just the people around me. Many others are also troubled by wanting to travel but not having the time.

Interview

What would be needed to recreate the sensory experience of your favorite scenic places at home?



Emily
20 years old
College student

I want to enjoy the scenery at home to save time, a panoramic view with a breeze would be better.



Tim
19 years old
High school student

I don't have a specific scenery I want to see, but it would be great to feel the real scenery at home, with people moving around.



Coley
24 years old
Office worker

I wish to revisit many places, would love to enjoy the scenery at home, accompanied by friends, with corresponding scents, and touch would be even better.



Jerry
26 years old
Teacher

I want to visit New York and Tibet, experiencing the natural scenery at home is very convenient, vision is key, multi-sensory experience, and it could be especially helpful for people with disabilities.

Summary

1. Busy schedules hinder relaxation and travel.
2. Desire for scenery to ease the mind is often unmet.

Persona



Jackson

DEMOOGRAPHIC

Age: 17
Location: Beijing
Gender: Male
Occupation: High School Student

NEEDS

He wants to experience the atmosphere of travel at home and see beautiful scenery at any time.

PAINPOINTS

He has neither the time nor the money to travel.

CHARACTERISTICS

He has many places he wants to visit and loves traveling, but he doesn't have money and is under significant academic pressure.



Tina

DEMOOGRAPHIC

Age: 28
Location: Shanghai
Gender: Female
Occupation: Office Worker

NEEDS

She wants a work or relaxation environment filled with the atmosphere of travel and a way to reminisce about her wonderful travel experiences.

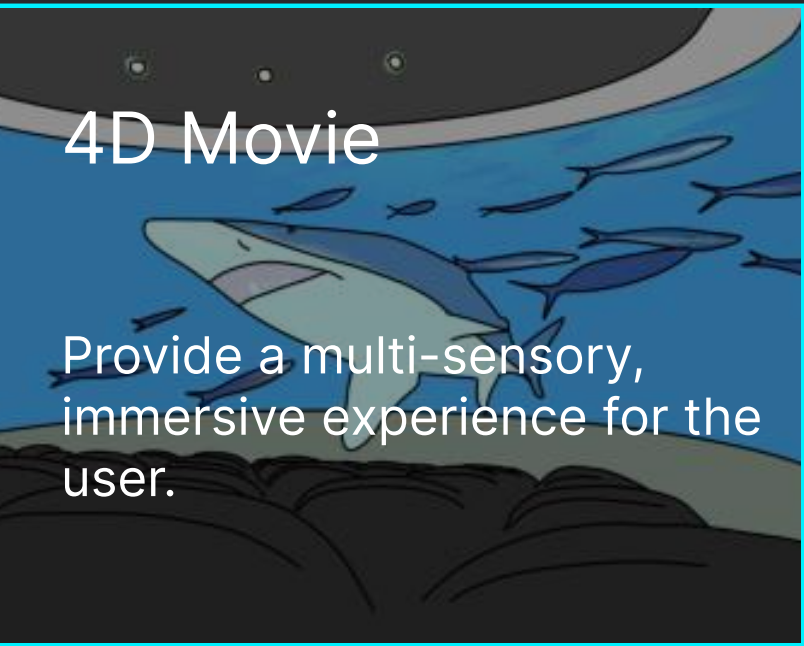
PAINPOINTS

Her busy work schedule makes it difficult to travel frequently to different places.

CHARACTERISTICS

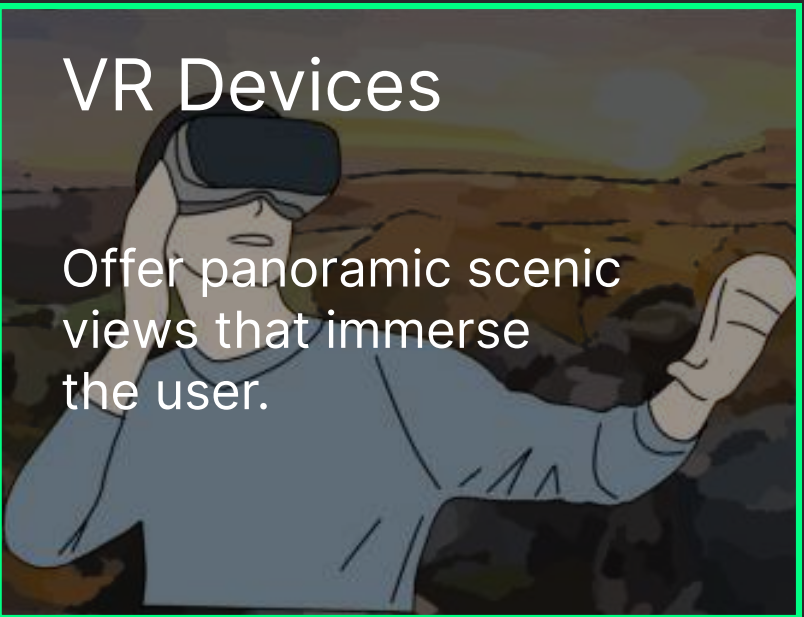
She loves traveling and has been to many different places. However, her work is now very busy, and she has fewer opportunities to travel.

Brain storm



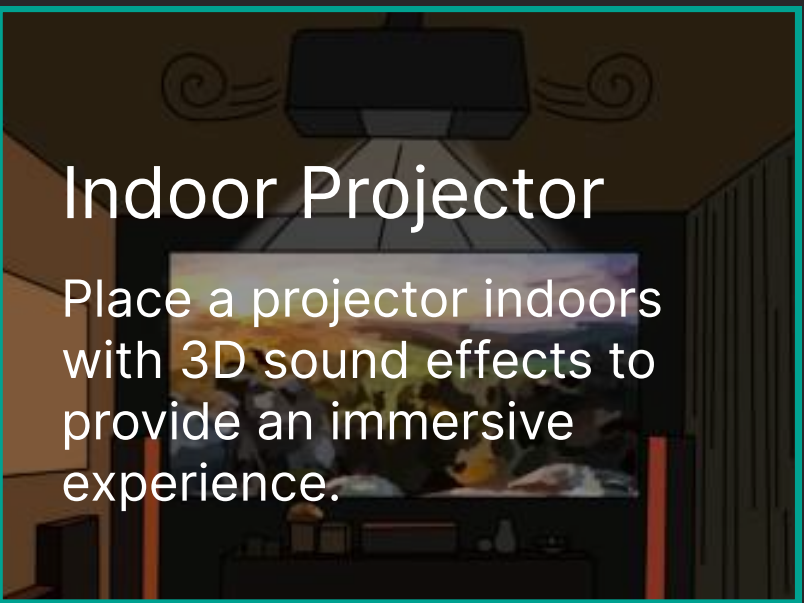
4D Movie

Provide a multi-sensory, immersive experience for the user.



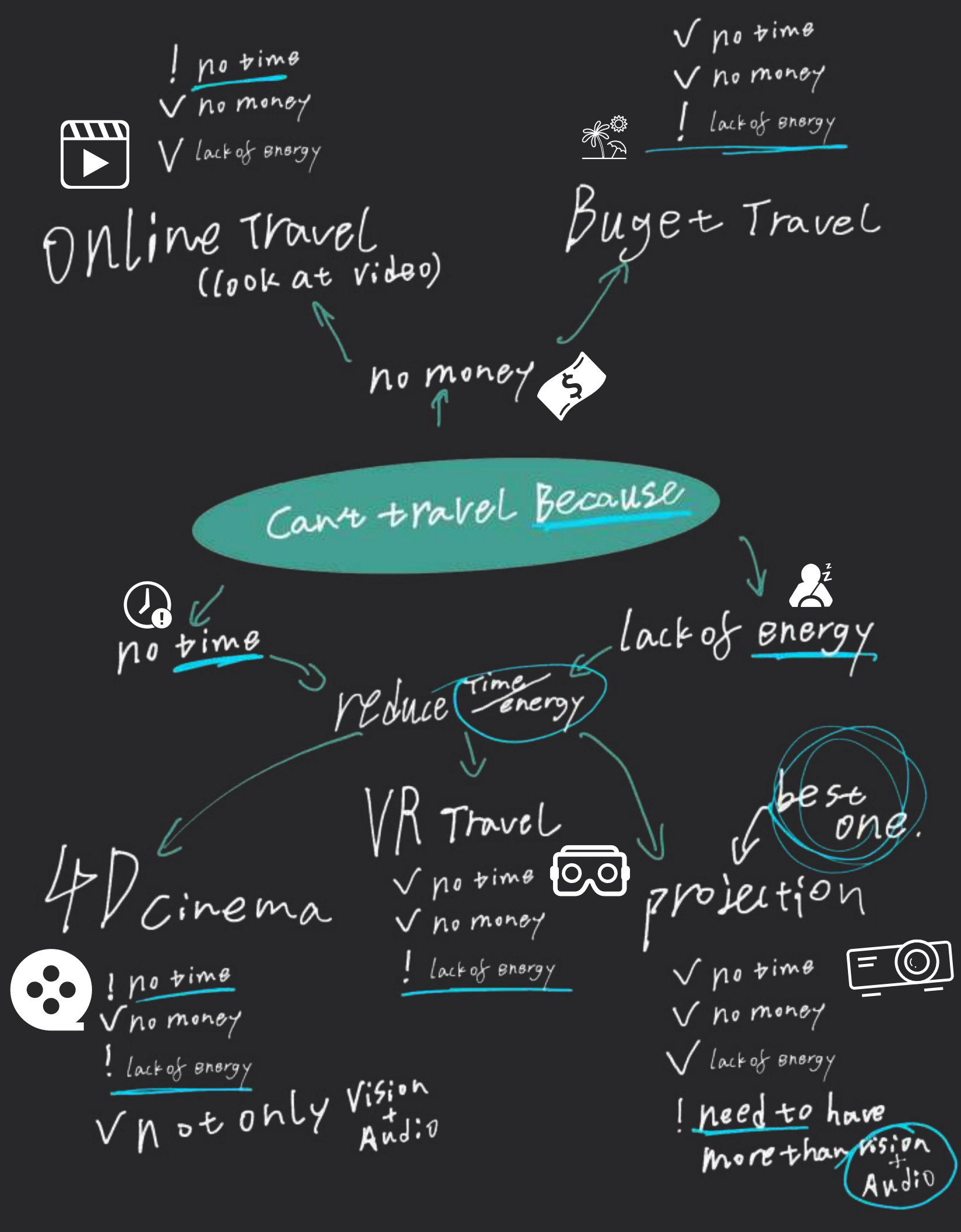
VR Devices

Offer panoramic scenic views that immerse the user.



Indoor Projector

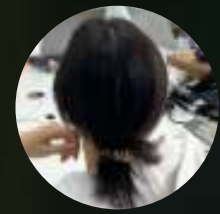
Place a projector indoors with 3D sound effects to provide an immersive experience.



Workshop



I designed and outlined the general direction of my four feasible proposals on a blackboard, and I found some friends to share and explain my proposals to them in order to gather their feedback.



4D Movie has strong immersion and multitasking but is costly and requires special equipment.

VR Devices are fully immersive but don't allow multitasking.



Indoor Projector is immersive but needs a specific device and lacks strong immersion.

I think I can combine the advantages of these features and create a modular design.

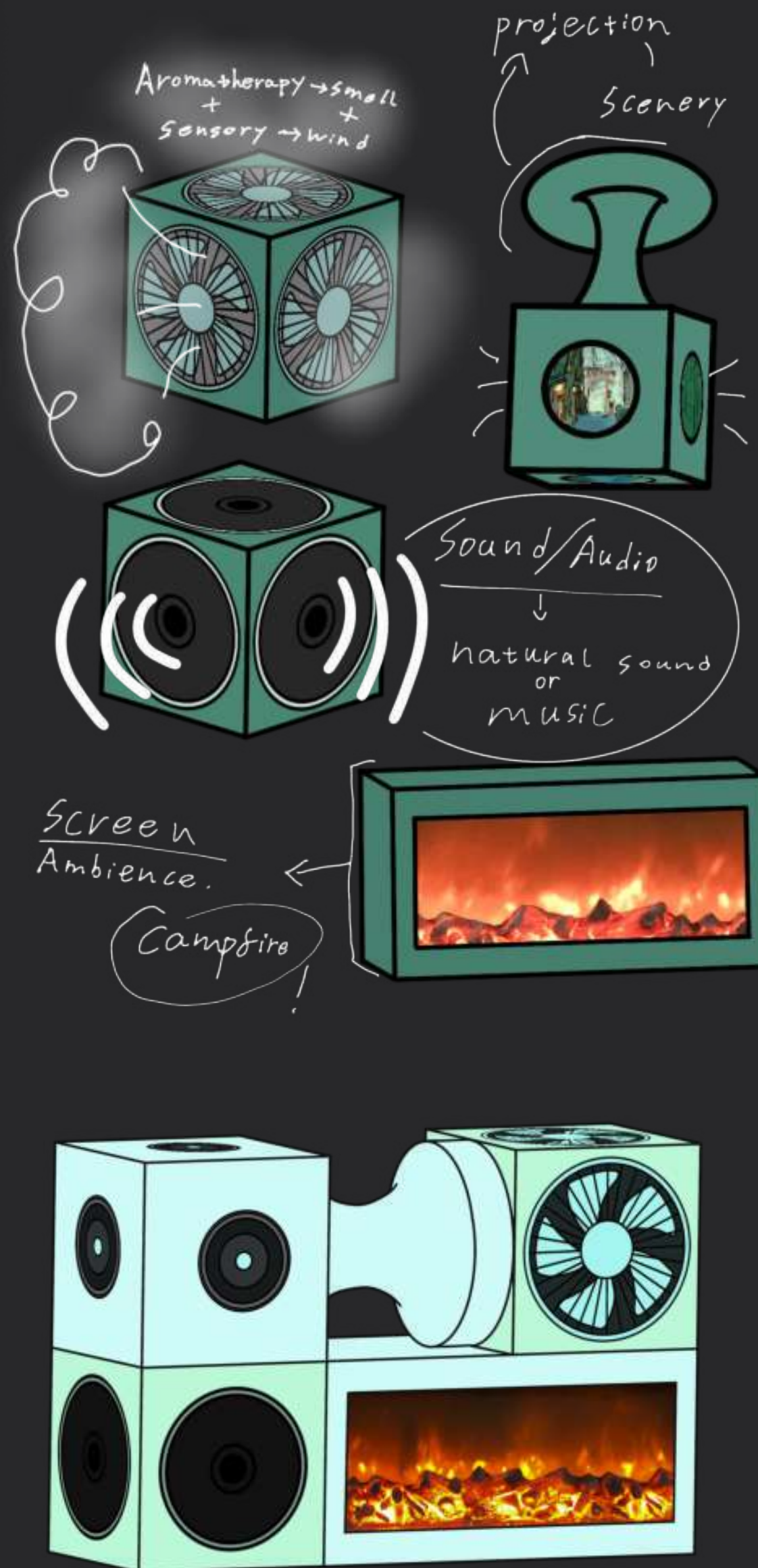


A ceiling-mounted panoramic projector covering the whole room in color would be awesome.

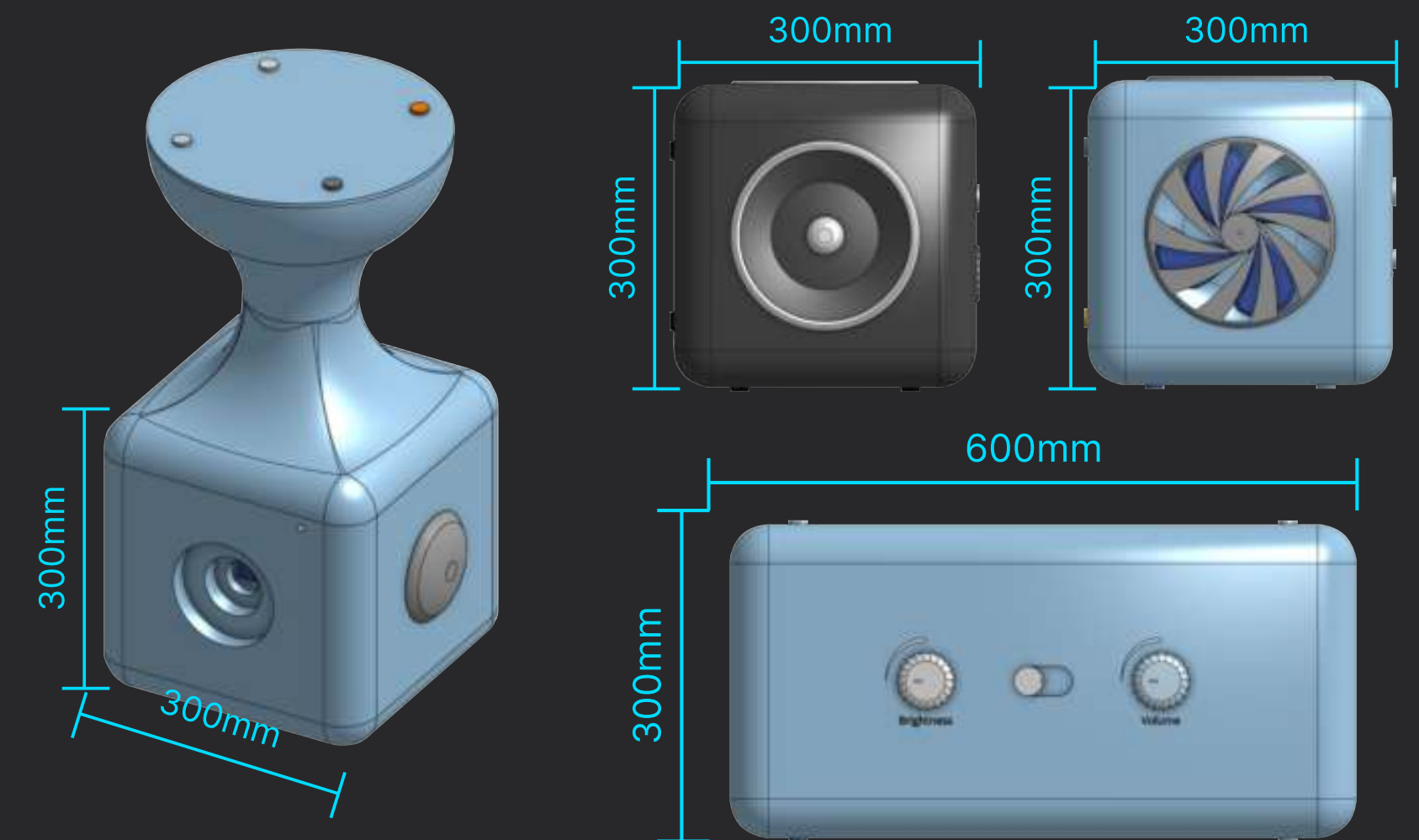
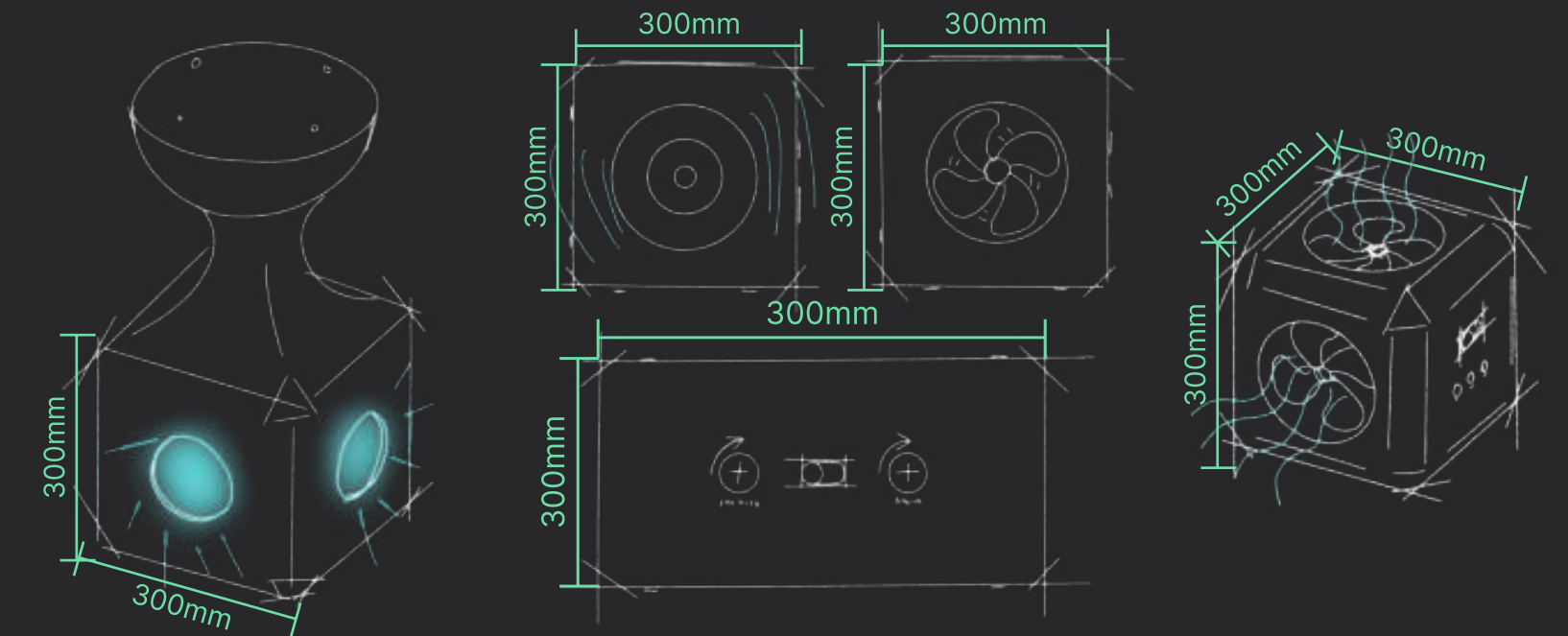
Conclusion

The initial four plans each have their own advantages and disadvantages. I have decided to combine their strengths and divide them into smaller modules, allowing for customization to meet everyone's needs. People can choose the modules they prefer based on their own preferences.

Concept



I designed a set of modular products that simulate the feeling of being outdoors through components like speakers, projectors, fans, and ambient lights. These components can be assembled when not in use to save storage space.

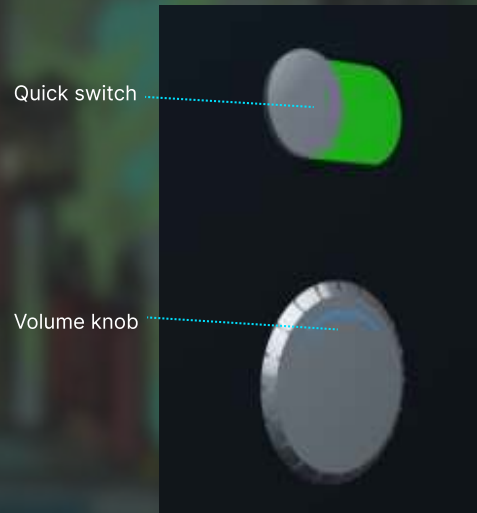


Product functions



Audio Module

Can adjust background sound effects and volume via a mobile app. The module also features a volume adjustment knob and a power switch.



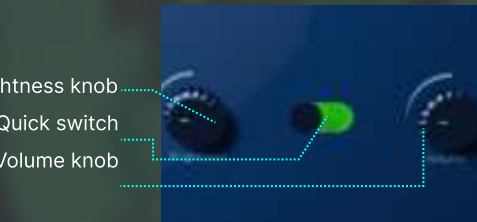
Aromatherapy Module

Built-in multiple natural scents, adjustable via a mobile app or customizable. Includes a fan for quick diffusion of scents, creating a real natural feel. Equipped with a quick adjustment knob for three preset scents and a quick switch.



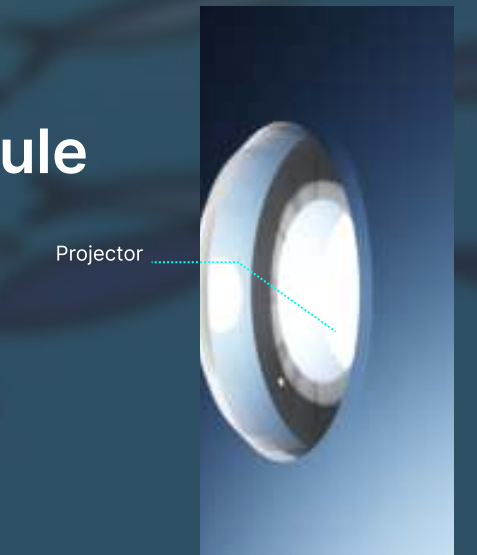
Ambient Display Module

Allows users to upload images or videos via mobile, which the module will automatically loop. Includes brightness and volume adjustment knobs and a quick switch for personalized comfort.

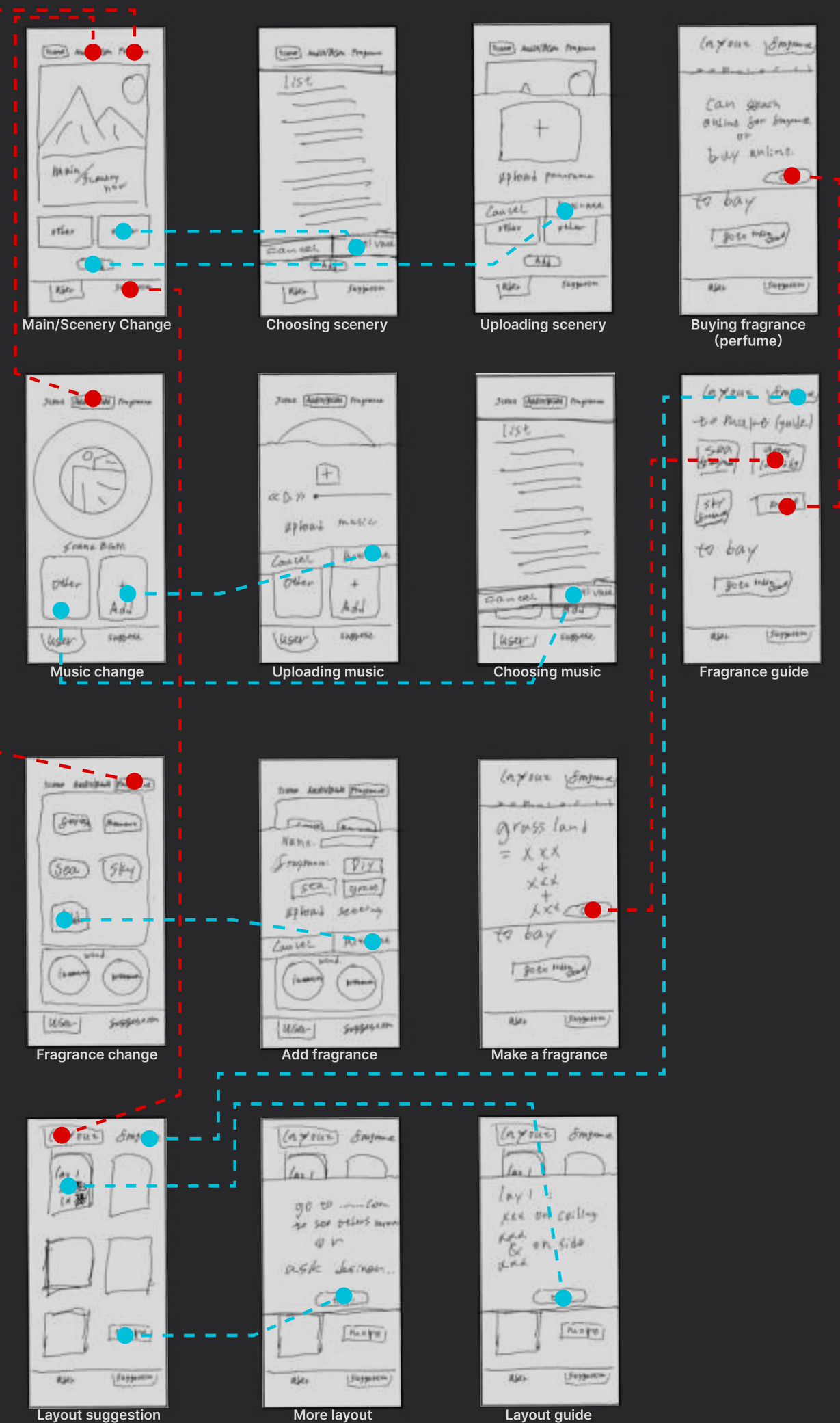


Scenery Simulation Module

360-degree panoramic projection for an immersive experience. Users can upload panoramic photos/videos via the mobile app or use recommended videos from the app.



Wireframe



The app can be used to adjust the settings for each module, such as the projected images and brightness, the direction and speed of the aroma diffuser, the fragrance, the display of ambient lighting images, music playback from the speakers, and more. In addition, the app provides various placement suggestions for the product and tutorials on how to create fragrances.

Usertest

STEP1

High school students

Simulated testing

I found a high school classmate to test the flow of my app and identified some issues in the app's process.

STEP2

Collect feedback

She felt that some parts of the app interface were unclear on how to operate and provided me with some suggestions for improvement.

Office workers

STEP1

Simulated testing

I found a working professional to test the flow of my app and identified some issues with the app's interface design.

STEP2

Collect feedback

He felt that certain aspects of the interface design had flaws and provided some better design solutions.

Development

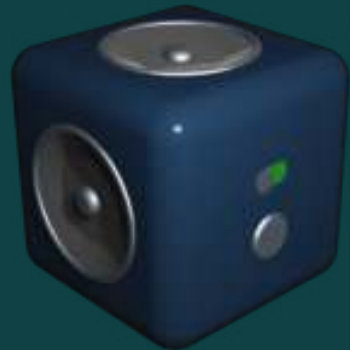


Interface Design



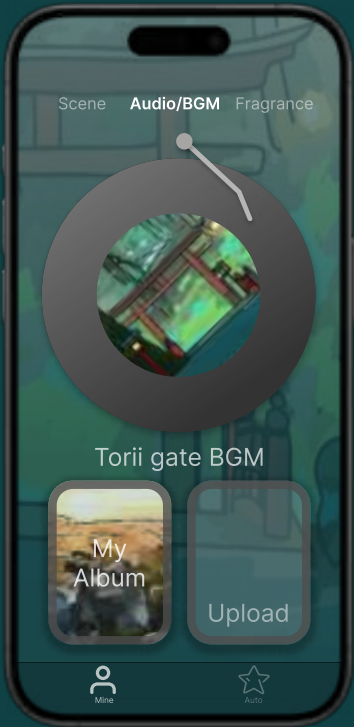
Projector control

This page is used to display the current projector scene and control the projector to change scenes.



Speaker control

This page is used to display the current music playing on the speakers and control the speakers to change the music.



Haptic/Fragrance Control

This section is used to display and control the direction and the intensity of airflow and the fragrance of the diffuser.

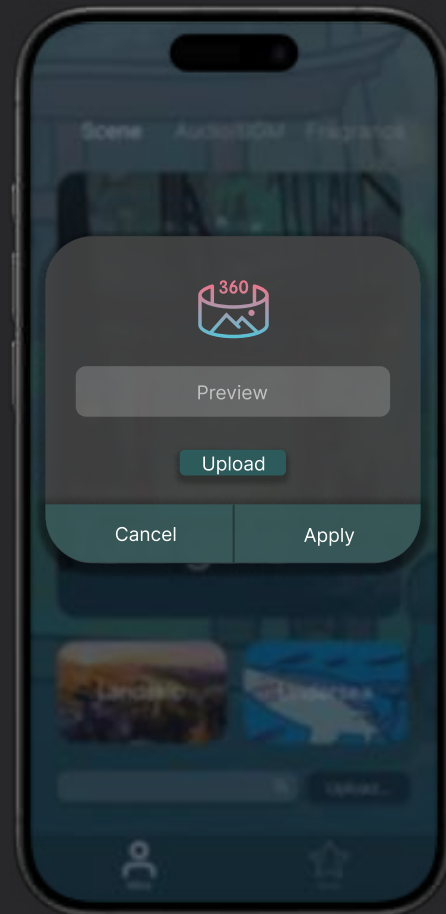
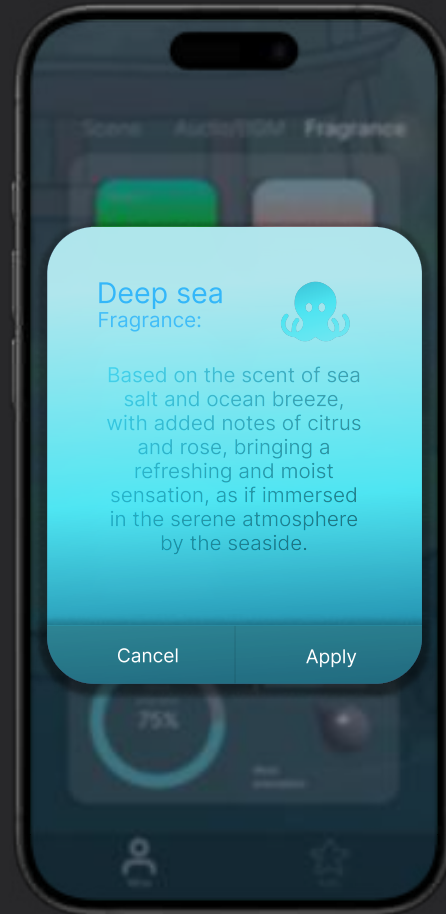
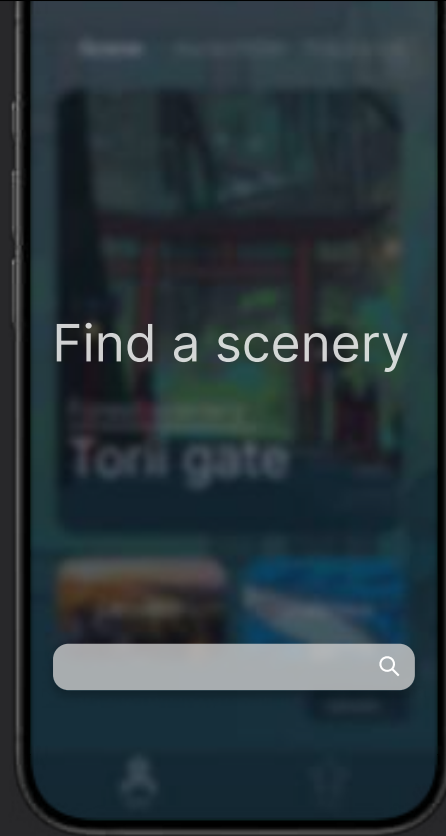
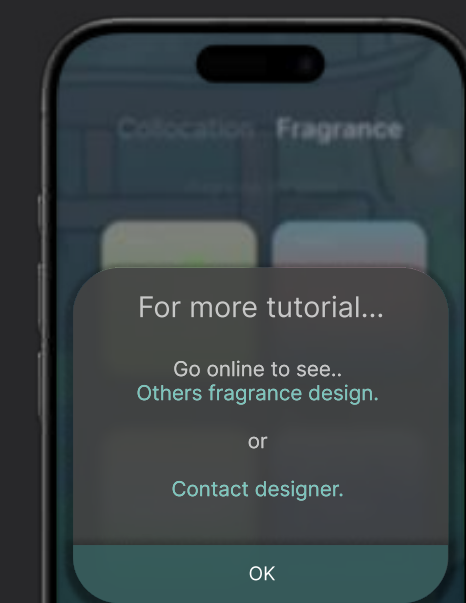
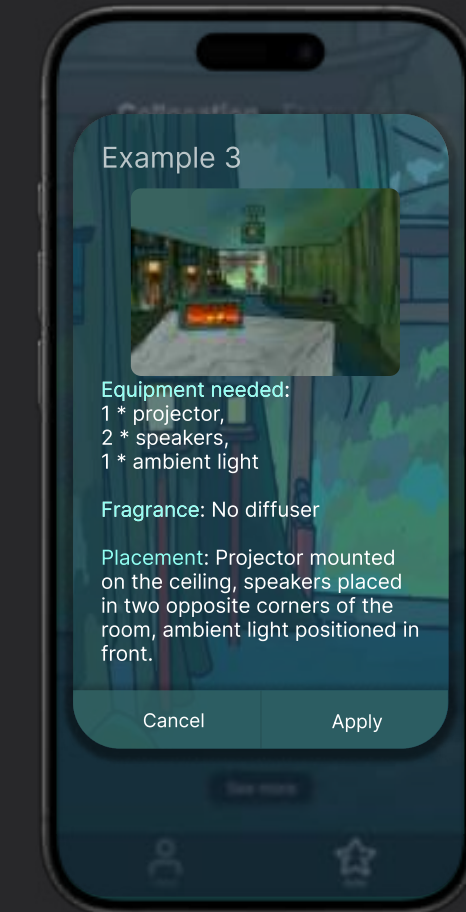
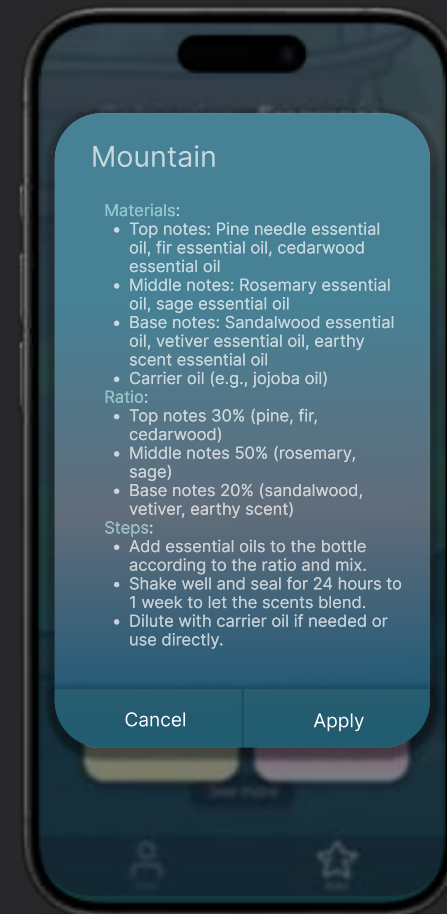
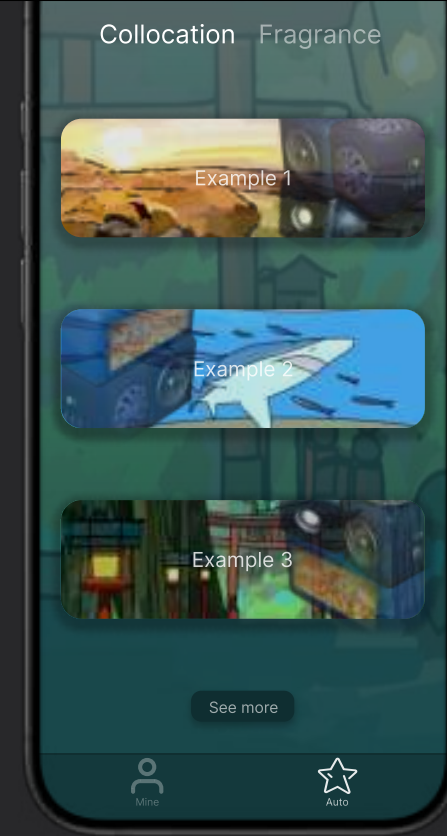
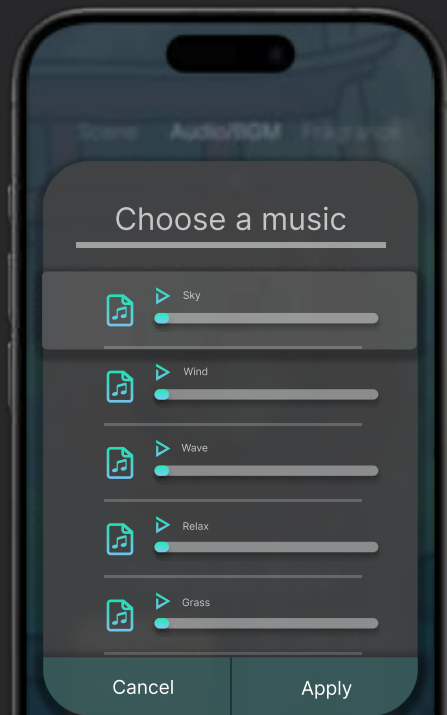
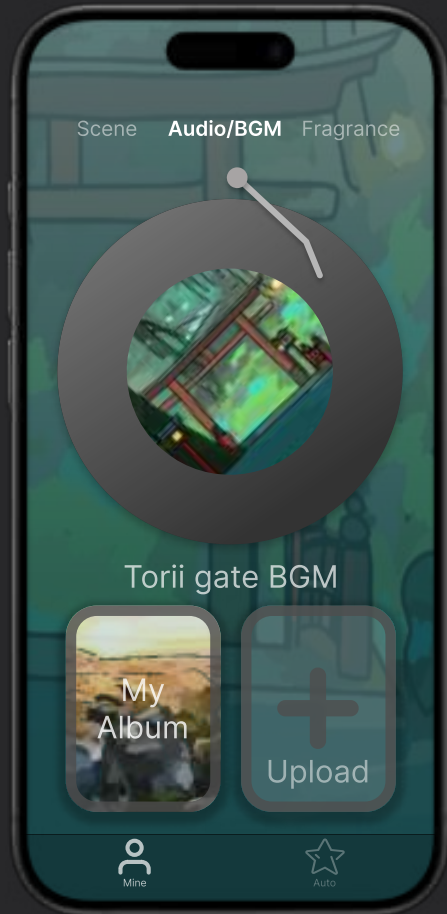
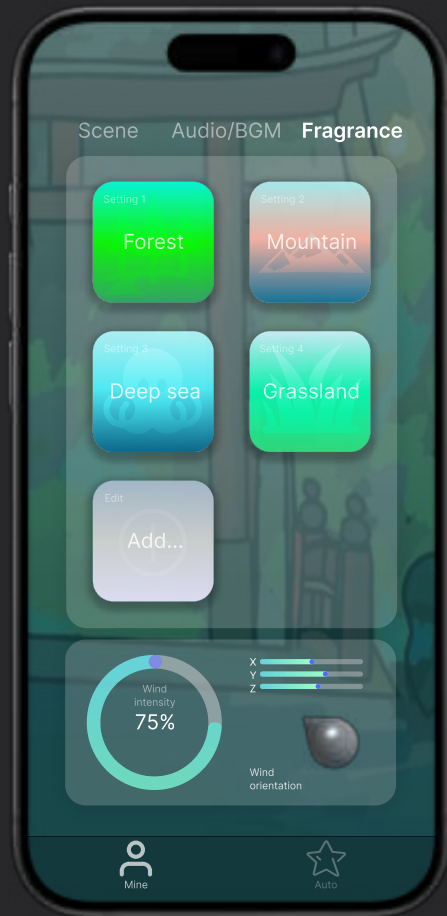
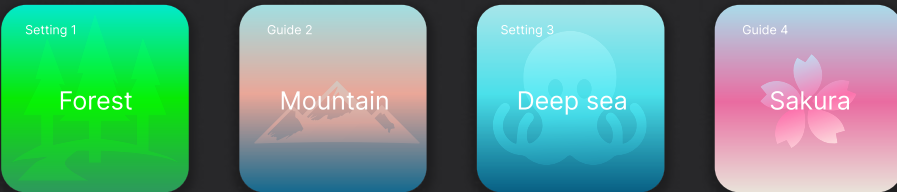
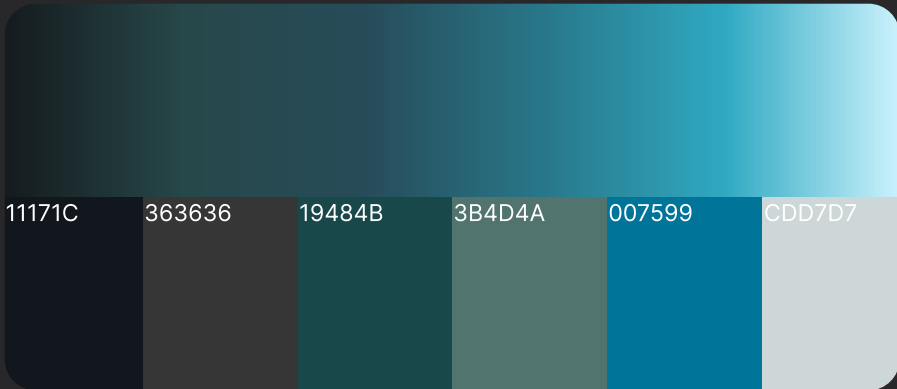
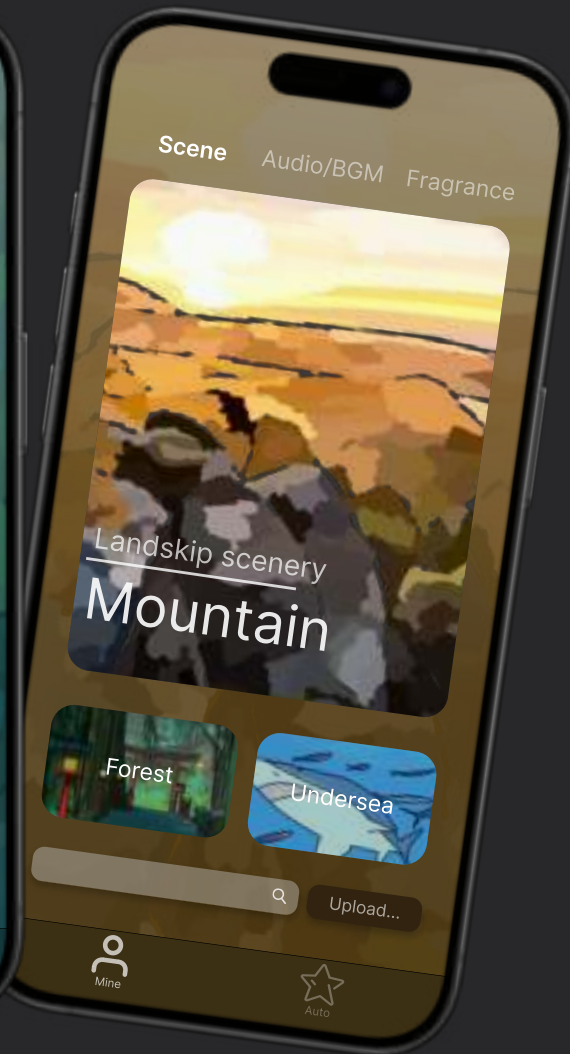


Configuration Suggestions

If users have no ideas on how to arrange or set up a scene, they can use system suggestions to directly apply presets.



UI Design



Scenarios



According to preferences:
All components can be customized in placement, allowing for flexible arrangement based on the home environment or personal preferences.



Travel the world from home:
Projectors, speakers, and ambient lighting create a warm and immersive indoor environment.



Change the music settings:
You can customize music, fragrances, ambient lighting, and more using the mobile app.



Change the environment settings:
Use a mobile app to customize shortcuts for creating the most comfortable background tailored to your needs.

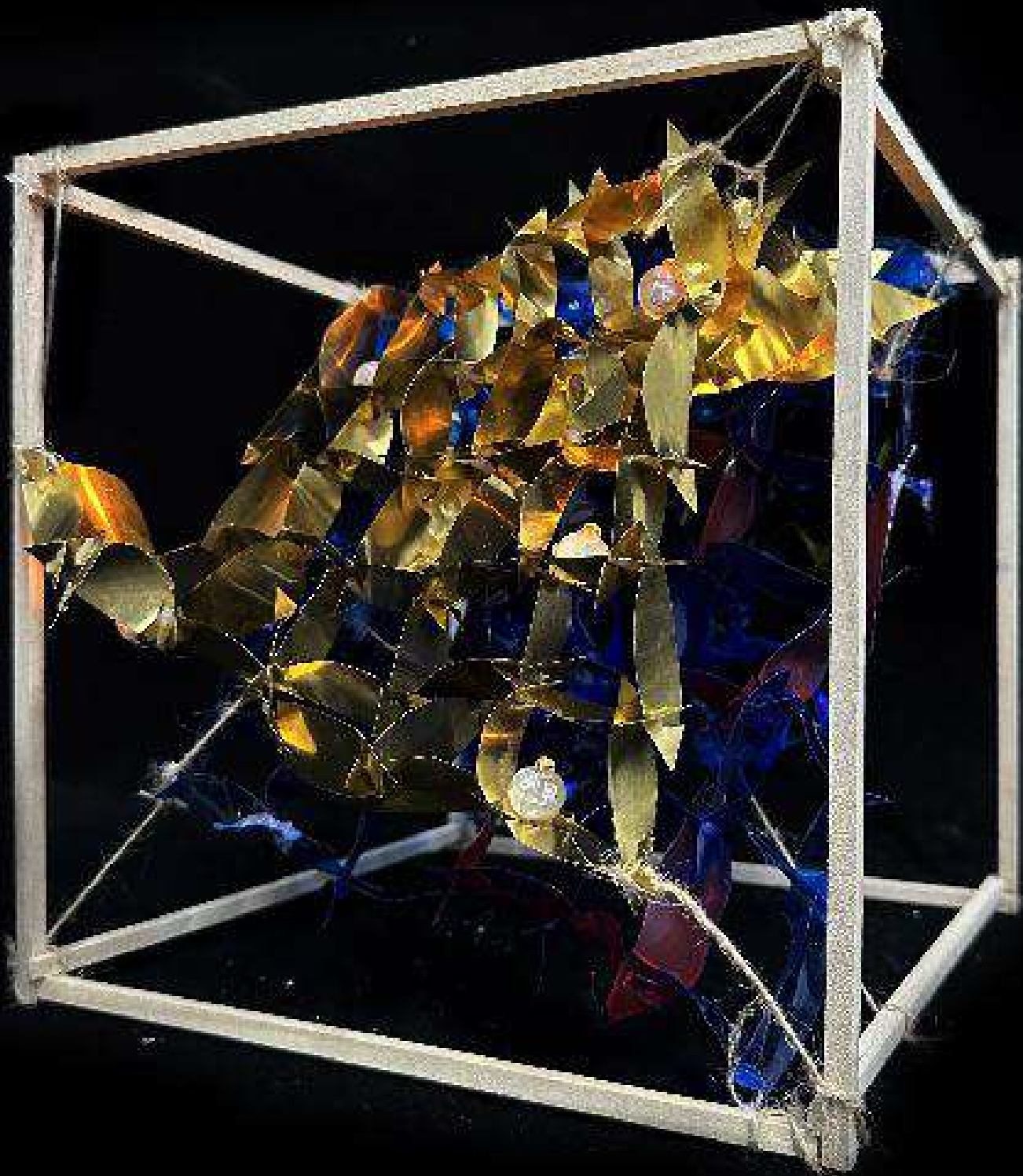
Storyboard



CRYSTALLIZATION SIMULATION

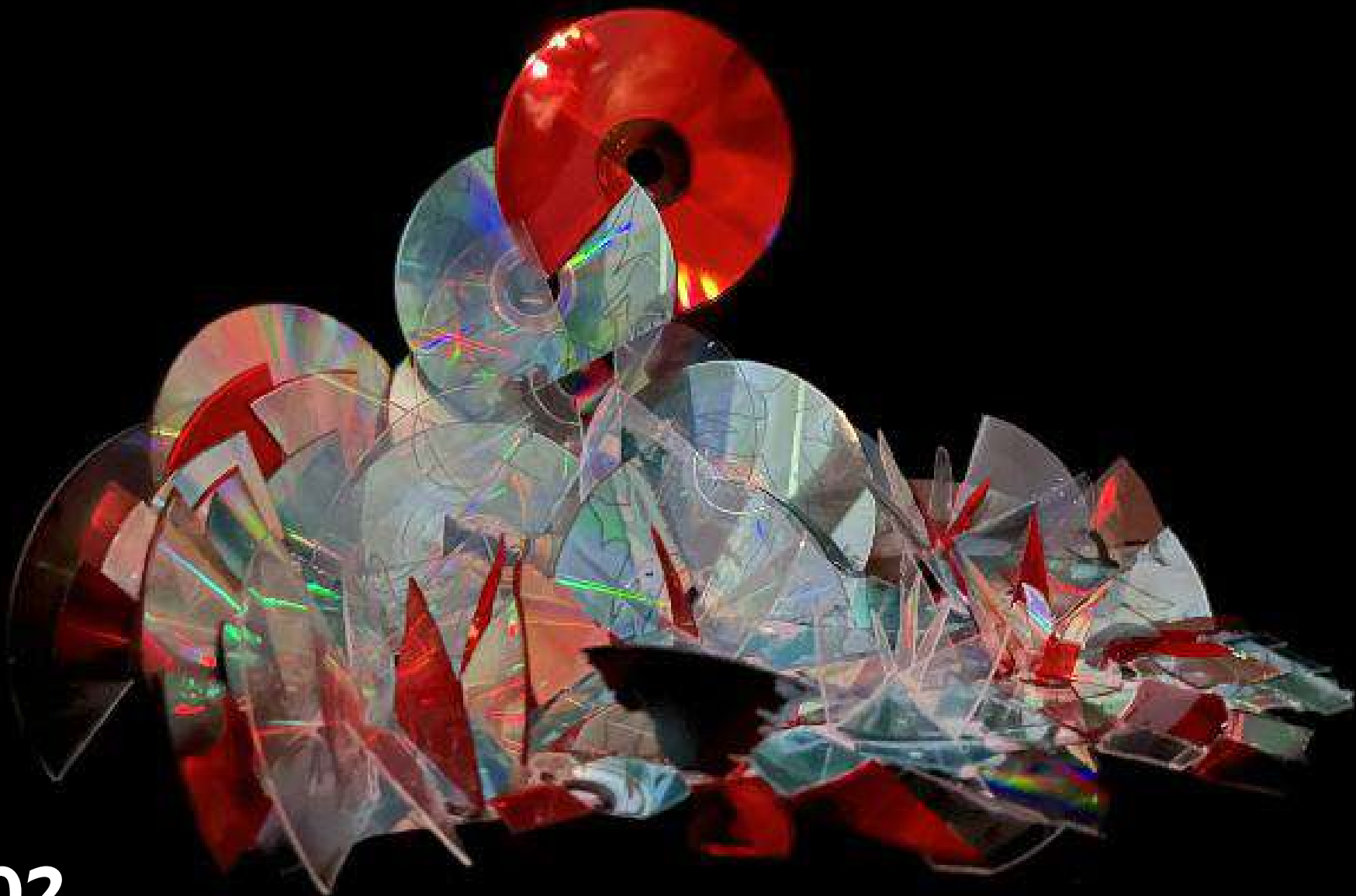
01 CRYSTALLINE UNIT

Simulating the Condensation Process of a Single Crystal Unit
Size: 500mm*500mm*500mm
Materials used: Metal sheets, wooden strips, decorative balls, hemp rope.



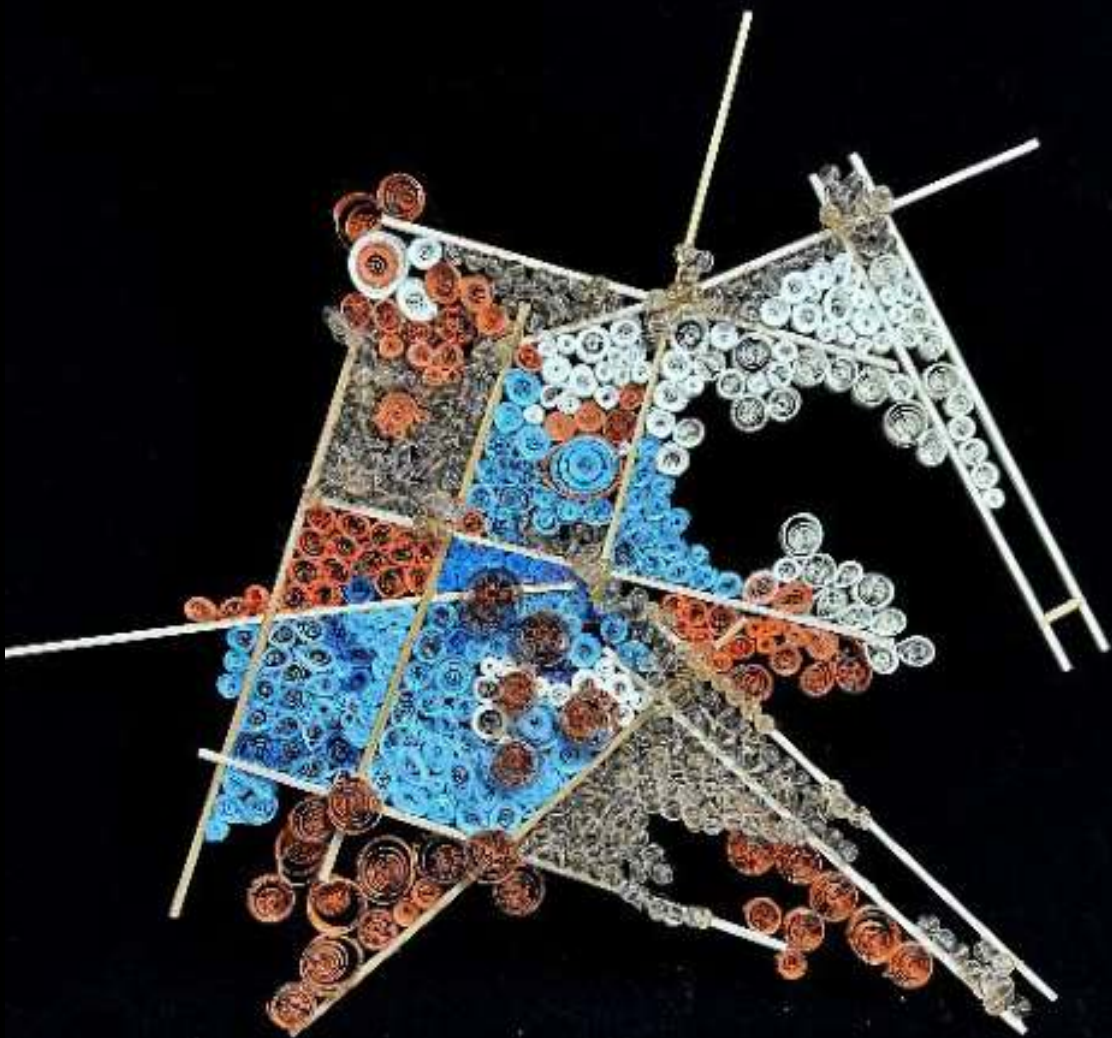
03 OCEANIC CRYSTALLIZATION

Simulating Beautiful Crystals in the Ocean
Size: 500mm*500mm*200mm
Materials used: Fishing line, aluminum foil, wooden strips, plastic beads.



02 TERRESTRIAL CRYSTALLIZATION

Simulating the Crystallization (Grow) of Terrestrial Life
Size: 500mm*500mm*400mm
Materials used: Disc, glass, marker, transparent cardstock



04 AERIAL CRYSTALLIZATION

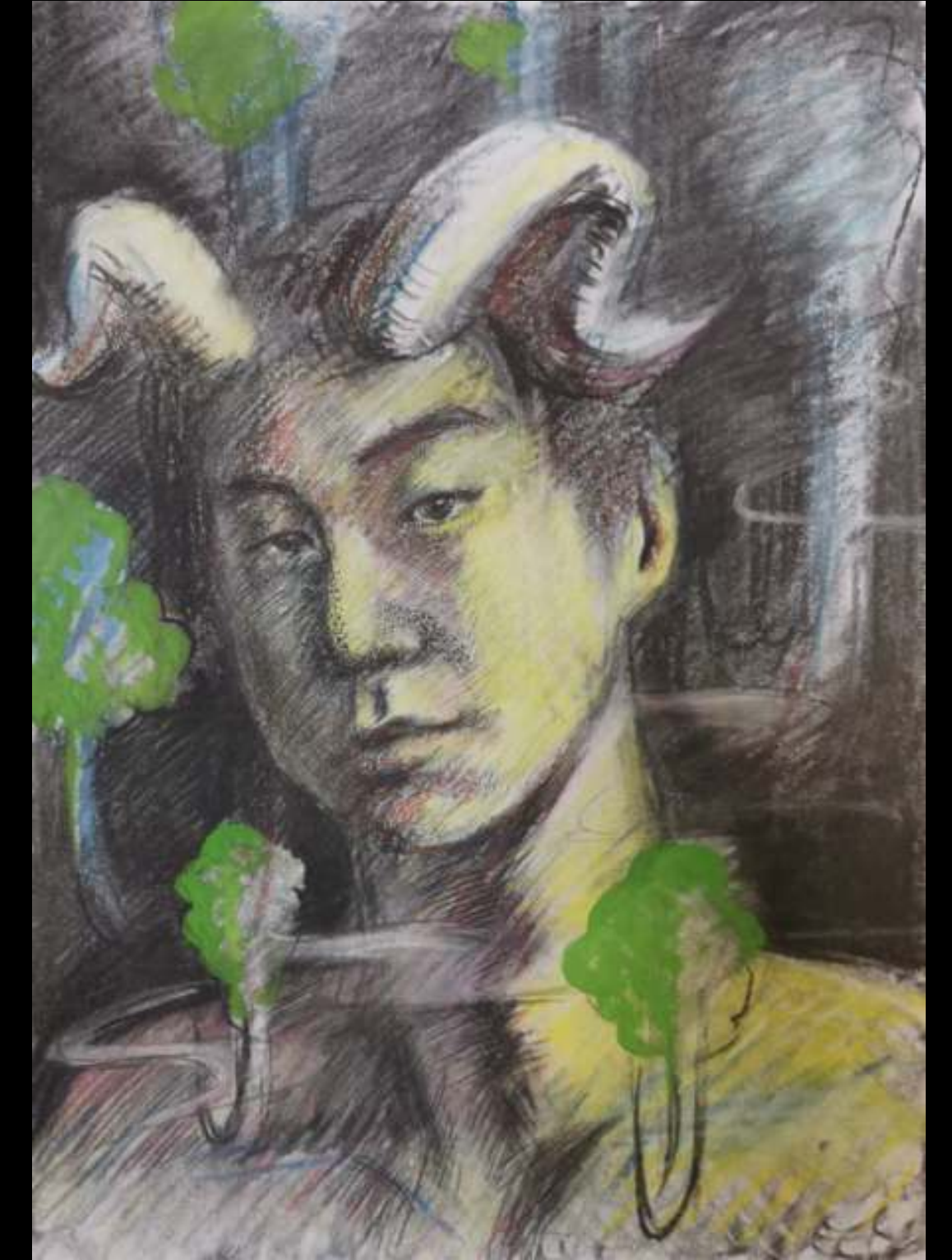
Simulating the Condensation of Water Vapor in the Sky
Size: 600mm*500mm*10mm
Materials used: Wooden strips, colored cardstock



Title: Chrono-Architecture: The Evolution of Form
*Size: 220mm*297mm*
Materials used: Printer, acrylics, outliner, pencil, scissors
Year: 2024



Title: Emotion Restraint: The Visage of Fury
*Size: 210mm*297mm*
Materials used: Charcoal pencils, pastels, colored pencils
Year: 2023



Title: Astral Self-Portrait: Aries
*Size: 210mm*297mm*
Materials used: Charcoal pencils, pastels, colored pencils
Year: 2023